CLINICAL CONDITIONS

Neurology Practice Tests

3 tests of 50 USMLE-style questions, with answers and explanations

For educational use only. This is a study aid, not a clinical reference.

Instructions

60 minutes	50 questions	About 72 seconds
for each test	in each test	for each question

Taking a test

- Take each test in one sitting, with a timer set to 60 minutes. Do not pause the timer.
- Each question has five options and one best answer. Choose the single best option, even if more than one seems reasonable.
- Mark your answers on the answer sheet after the last question of each test, or on a separate piece of paper.
- Do not use notes, the website, or other references. The abbreviation list on the next page is allowed.
- There is no penalty for guessing, so answer every question. An unanswered question counts as incorrect.
- When time ends, stop, even if you have not finished.

About the tests

- No question appears in more than one test, so the three tests can be taken in any order.
- Every test has 15 physiology and 35 clinical questions, covering all 12 physiology topics and all 23 clinical topics, with a similar mix of Step 1 questions (mechanisms, microbiology, and pathology) and Step 2 questions (diagnosis, next step, and management).
- The tests use 150 of the 225 questions on the website, so the website quiz has more practice.
- Questions are in random order, and the topic being tested is not shown until the answers.

Scoring and review

- Your score is the number of correct answers. Use the answer key at the start of each answer section, and write your score on your answer sheet.
- Then read the explanations. Each one names the topic and step, so you can see what to revisit.
- Answers follow the course material. Where a guideline differs, the explanation says so under "Additional information".

Contents

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Test A answers
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Abbreviations

You may refer to this page during a test.

ACA Anterior cerebral artery

ACE Angiotensin-converting enzyme

ALS Amyotrophic lateral sclerosis

ANA Antinuclear antibody

ANP Atrial natriuretic peptide

APOE Apolipoprotein E

APP Amyloid precursor protein

AQP4 Aquaporin-4

AQP4-IgG Aquaporin-4 immunoglobulin G antibody

AREDS Age-Related Eye Disease Study

Aβ42 Amyloid beta 42, the 42-amino-acid form

B12 Vitamin B12 (cobalamin)

BBB Blood-brain barrier

BMI Body mass index

BP Blood pressure

BPPV Benign paroxysmal positional vertigo

bvFTD Behavioral variant frontotemporal dementia

C9orf72 Chromosome 9 open reading frame 72 gene

CAG Cytosine-adenine-guanine trinucleotide

CMV Cytomegalovirus

CN Cranial nerve

CN III Cranial nerve III (oculomotor)

CN VI Cranial nerve VI (abducens)

CN VII Cranial nerve VII (facial)

CN X Cranial nerve X (the vagus nerve)

CNS Central nervous system

COMT Catechol-O-methyltransferase

COPD Chronic obstructive pulmonary disease

CPP Cerebral perfusion pressure

CSF Cerebrospinal fluid

CT Computed tomography

CVLM Caudal ventrolateral medulla

CXR Chest x-ray

D1 Dopamine receptor 1

D2 Dopamine receptor 2

DAT Dopamine transporter

DLB Dementia with Lewy bodies

DOAC Direct oral anticoagulant

ECG Electrocardiogram

EDNRB Endothelin receptor type B gene

EEG Electroencephalogram

ESR Erythrocyte sedimentation rate

G6PD Glucose-6-phosphate dehydrogenase

GCS Glasgow Coma Scale

GJB2 Gap junction beta 2 gene (connexin 26)

GPe Globus pallidus externus

GPi Globus pallidus internus

HLA-B*1502 Human leukocyte antigen B*15:02 allele

HLA-B27 Human leukocyte antigen B27

HR Heart rate

HTT Huntingtin gene

ICA Internal carotid artery

ICH Intracerebral hemorrhage

ICP Intracranial pressure

INR International normalized ratio

IOL Intraocular lens

IOP Intraocular pressure

IV Intravenous

IVIG Intravenous immunoglobulin

LVO Large vessel occlusion

MAO-B Monoamine oxidase B

MAPT Microtubule-associated protein tau gene

MCA Middle cerebral artery

MI Myocardial infarction

MITF Melanocyte-inducing transcription factor gene

MLF Medial longitudinal fasciculus

MMSE Mini-Mental State Examination

MOG-IgG Myelin oligodendrocyte glycoprotein immunoglobulin G antibody

MOGAD MOG antibody-associated disease

MRA Magnetic resonance angiography

MRI Magnetic resonance imaging

MS Multiple sclerosis

MSA Multiple system atrophy

NMDA N-methyl-D-aspartate

NMOSD Neuromyelitis optica spectrum disorder

NSF Nephrogenic systemic fibrosis

NTS Nucleus tractus solitarius

OCP Oral contraceptive pill

OCT Optical coherence tomography

PaCO2 Arterial partial pressure of carbon dioxide

PAX3 Paired box 3 gene

PET Positron emission tomography

PNES Psychogenic nonepileptic seizures

PSEN1 Presenilin 1

PSEN2 Presenilin 2

RAPD Relative afferent pupillary defect

RPE Retinal pigment epithelium

RVLM Rostral ventrolateral medulla

S1 Primary somatosensory cortex

SAH Subarachnoid hemorrhage

SNc Substantia nigra pars compacta

SNHL Sensorineural hearing loss

SPECT Single-photon emission computed tomography

SSRI Selective serotonin reuptake inhibitor

STN Subthalamic nucleus

SWI Susceptibility-weighted imaging

TFTs Thyroid function tests

TPMT Thiopurine methyltransferase

VA Ventral anterior nucleus of the thalamus

VEGF Vascular endothelial growth factor

VL Ventral lateral nucleus of the thalamus

VMAT2 Vesicular monoamine transporter 2

Test A

50 questions · 60 minutes

1. An 84-year-old woman in a nursing home became confused over 2 days. Her attention wanders, and she is drowsy in the morning but agitated at night. She has a fever and cloudy urine.

Which is the most likely cause of her confusion?

A. Alzheimer disease
B. Frontotemporal dementia
C. Vascular dementia
D. Delirium
E. Normal pressure hydrocephalus

2. A 38-year-old woman has severe bilateral optic neuritis and a cord lesion spanning 5 vertebral segments. Serum AQP4-IgG is positive.

Which cell is the primary target of the disease?

A. Motor neuron
B. Astrocyte
C. Oligodendrocyte
D. Schwann cell
E. Microglia

3. An 80-year-old woman with a long-neglected, swollen white cataract in her right eye develops sudden pain and redness in that eye. IOP is 46 mmHg, and gonioscopy shows a closed angle.

Which is the mechanism and treatment?

A. Open-angle glaucoma; latanoprost
B. Phacomorphic glaucoma; lens removal
C. Bacterial keratitis; topical antibiotics
D. Anterior uveitis; topical steroids
E. Phacolytic glaucoma; lens removal

4. A man with two previous strokes, one in each hemisphere, has slurred speech, difficulty swallowing, a brisk jaw jerk, and no tongue wasting.

Which lesion explains this?

A. A lesion of the hypoglossal nucleus
B. A lesion of the nucleus ambiguus
C. A cerebellar lesion
D. Bilateral corticobulbar lesions
E. A unilateral corticospinal lesion

5. A man stubs his toe and feels an immediate sharp, well-localized pain, followed a second later by a dull, spreading ache.

Which fibers carry the second, dull pain?

A. Unmyelinated C fibers
B. Large myelinated Aβ fibers
C. Aα motor fibers
D. Lightly myelinated Aδ fibers
E. Preganglionic sympathetic fibers

6. A 44-year-old woman had a sudden, severe headache that peaked within a minute while lifting weights 10 hours ago. She is alert with mild neck stiffness. Non-contrast CT head is normal.

Which is the most appropriate next step?

A. Lumbar puncture for xanthochromia
B. MRI of the cervical spine
C. EEG
D. Discharge with analgesia
E. Start nimodipine and repeat CT in a week

7. A 66-year-old man has a stiff, jerky right arm that sometimes rises on its own as though controlled by someone else. He cannot copy hand gestures with it or identify a coin placed in that hand with his eyes closed.

Which diagnosis is most likely?

A. Multiple system atrophy
B. Huntington disease
C. Corticobasal degeneration
D. Progressive supranuclear palsy
E. Essential tremor

8. A 30-year-old teacher has a red, watery, itchy left eye a week after a cold, now spreading to the right eye. Vision is normal, and she has a tender preauricular node.

Which is the most appropriate management?

A. Artificial tears, cool compresses, and hand hygiene
B. Topical corticosteroids
C. Oral amoxicillin
D. Urgent Gram stain and culture
E. Topical chloramphenicol

9. A man has total deafness in his right ear after a stroke.

Why is a cortical stroke unlikely to be the cause?

A. The cochlear nerve does not reach the brainstem
B. Hearing becomes bilateral above the cochlear nuclei
C. The auditory cortex lies in the occipital lobe
D. The auditory cortex receives input only from the opposite ear
E. Hearing is carried by CN VII

10. A 62-year-old smoker has proximal leg weakness, dry mouth, and reduced reflexes that briefly improve after repeated contraction. CXR shows a hilar mass.

What is the target of the autoantibody?

A. Acetylcholinesterase
B. Voltage-gated sodium channels
C. Presynaptic voltage-gated calcium channels
D. Muscle-specific kinase
E. Postsynaptic nicotinic acetylcholine receptors

11. A 26-year-old woman with highly active relapsing MS is to start natalizumab.

Which test should be checked first?

A. AQP4 antibody
B. Thiopurine methyltransferase
C. HLA-B*1502
D. JC virus status
E. First-dose ECG monitoring

12. A 72-year-old woman developed symmetric slowness and rigidity of both arms 2 months after starting a drug for gastroparesis. She has no tremor at rest.

Which drug is most likely responsible?

A. Metoclopramide
B. Erythromycin
C. Ondansetron
D. Lactulose
E. Omeprazole

13. A pathologist stains CNS tissue for the cell that buffers extracellular potassium, takes up neurotransmitter, and contributes foot processes to the BBB.

Which cell is this?

A. Microglia
B. Ependymal cell
C. Oligodendrocyte
D. Schwann cell
E. Astrocyte

14. A 41-year-old woman with AQP4-IgG-positive NMOSD has recovered from her first attack of myelitis.

Which long-term treatment should be avoided?

A. Rituximab
B. Satralizumab
C. Eculizumab after meningococcal vaccination
D. Azathioprine with prednisone
E. Interferon beta

15. At autopsy, a man who developed personality change at 55 has knife-edge atrophy of the frontal and anterior temporal lobes, with round tau-positive inclusions and ballooned neurons.

What are these inclusions?

A. Pick bodies
B. Glial cytoplasmic inclusions
C. Lewy bodies
D. Neurofibrillary tangles in the entorhinal cortex
E. Huntingtin nuclear inclusions

16. A 58-year-old man with 20 years of poorly controlled hypertension has a pontine hemorrhage.

Which vascular change most likely underlies the bleed?

A. Saccular aneurysm at a bifurcation of the circle of Willis
B. Charcot-Bouchard aneurysms in penetrating arterioles
C. A tangle of arteries draining directly into veins
D. Septic emboli lodging in distal arteries
E. Amyloid deposits in cortical vessel walls

17. A drug for chorea blocks the loading of dopamine into presynaptic vesicles.

Which drug is this?

A. Tetrabenazine
B. Rasagiline
C. Haloperidol
D. Pramipexole
E. Entacapone

18. After a stab wound to the right side of the back at T8, a man has weakness and loss of vibration and position sense in the right leg.

Where else is sensation lost?

A. All sensation in both legs
B. Pain and temperature in the left leg
C. Pain and temperature in both arms
D. Vibration in the left leg
E. Pain and temperature in the right leg

19. A 59-year-old smoker has right shoulder pain, mild ptosis, a small right pupil, and a dry right side of the face. CXR shows a right apical mass.

Which neurons are damaged?

A. Parasympathetic fibers in CN III
B. CN VII fibers to the levator
C. Edinger-Westphal neurons
D. Third-order neurons in the cavernous sinus only
E. Second-order sympathetic neurons leaving at T1

20. A 41-year-old man woke 2 days ago with severe hearing loss in his left ear. Otoscopy is normal, and audiometry shows a 50 dB sensorineural loss across 4 consecutive frequencies.

Which is the most appropriate treatment?

A. Oral amoxicillin
B. Oral prednisone 1 mg/kg for 7 days, then tapered
C. Microsuction
D. A hearing aid fitted once his hearing stabilizes in 3 months
E. Observation for 3 months

21. A student asks why patients with an acute left vestibular nerve lesion have nystagmus beating to the right.

Which statement explains this?

A. The eyes drift slowly toward the stronger side, and fast phases beat back toward the lesion
B. The cerebellum generates the fast phase toward the lesion
C. Fixation drives the eyes toward the side of the lesion
D. Nystagmus is named for its slow phase, which beats away from the lesion
E. The eyes drift slowly toward the weaker side, and fast phases beat away from it

22. A 59-year-old man with diabetes has burning pain in both feet that disturbs his sleep. He also has depression.

Which drug treats both, by strengthening descending inhibition?

A. Paracetamol
B. Ketamine
C. Ibuprofen
D. Duloxetine
E. Morphine

23. At autopsy, a patient with parkinsonism, early autonomic failure, and ataxia has α-synuclein inclusions in oligodendrocytes.

Which disease is this?

A. Typical Parkinson disease
B. Progressive supranuclear palsy
C. Multiple system atrophy
D. Corticobasal degeneration
E. Frontotemporal lobar degeneration

24. A 64-year-old man arrives 2 hours after sudden aphasia and right hemiparesis. CT head shows no hemorrhage. He takes warfarin for a mechanical valve, and his INR is 2.6. His BP is 170/95 mm Hg and platelets are 220000/μL.

Which finding contraindicates IV thrombolysis?

A. His BP
B. His INR
C. His time from onset
D. His aphasia
E. His mechanical valve

25. A 50-year-old woman with Huntington disease has chorea that makes her drop things and fall. She has a history of depression.

If tetrabenazine is started, what must be monitored?

A. Serum lithium level
B. Liver enzymes weekly
C. Mood and suicidal thoughts
D. Serum potassium
E. White cell count for agranulocytosis

26. A 76-year-old woman recovers well from a left MCA stroke. Telemetry shows A-fib, and echocardiogram shows a normal native mitral valve.

Which treatment best prevents recurrence?

A. Aspirin plus clopidogrel indefinitely
B. Carotid endarterectomy
C. Warfarin with a target INR of 3.0
D. A DOAC
E. Aspirin alone

27. A 34-year-old woman thought to have MS has severe optic neuritis in both eyes and a cord lesion spanning 6 vertebral segments.

Which test should be done before starting MS treatment?

A. AQP4-IgG
B. Serum ACE alone
C. DAT SPECT
D. Amyloid PET
E. EEG

28. A 35-year-old man has attacks of excruciating pain around his right eye lasting about an hour, with a watering eye and blocked nostril on that side. He paces during attacks, which wake him at night.

Which acute treatment is most effective?

A. Propranolol
B. 100% oxygen
C. High-dose prednisone
D. Lumbar puncture
E. Paracetamol

29. A tumor at the skull base causes numbness of the lower lip and chin and weakness of the masseter on one side.

Through which foramen does the affected division pass?

A. Jugular foramen
B. Foramen rotundum
C. Stylomastoid foramen
D. Foramen ovale
E. Superior orbital fissure

30. A 68-year-old man has a dense cataract in his left eye and reduced vision. The swinging flashlight test shows a left RAPD.

What does the RAPD indicate?

A. A posterior subcapsular cataract
B. Retinal or optic nerve disease in addition to the cataract
C. Phacolytic glaucoma
D. Nothing, since dense cataracts commonly cause an RAPD
E. The cataract alone is dense enough to cause the RAPD and needs surgery

31. A man cuts his median nerve at the wrist. A day later the distal axon and its myelin begin to break down, and the cell body swells with dispersed Nissl bodies.

What are these two processes?

A. Excitotoxicity and apoptosis
B. Segmental demyelination followed by remyelination
C. Axonal sprouting with pruning of distal synapses
D. Gliosis and neuronal loss
E. Wallerian degeneration and chromatolysis

32. A 70-year-old man with Parkinson disease starts levodopa combined with carbidopa.

What is the role of carbidopa?

A. It stimulates D3 receptors directly
B. It blocks NMDA receptors
C. It inhibits MAO-B in the brain
D. It inhibits COMT in the brain to prolong each dose
E. It blocks peripheral conversion of levodopa to dopamine

33. After a lymph node biopsy in the posterior triangle of the left neck, a woman cannot shrug her left shoulder well.

Which nerve was injured?

A. Left long thoracic nerve
B. Left CN XI
C. Left CN IX
D. Left CN X
E. Left CN XII

34. A 60-year-old man with cirrhosis is increasingly drowsy with asterixis. An EEG is obtained.

Which EEG finding is most likely?

A. A normal posterior alpha rhythm
B. Prominent posterior slow waves
C. Focal spikes over one temporal lobe
D. Generalized 3 Hz spike-and-wave
E. Triphasic waves

35. A 19-year-old man has had trouble seeing at night for years and now bumps into things at his sides. The fundus shows bone spicule pigment and narrowed arterioles.

Which cells are lost first?

A. Rods
B. RPE cells beneath drusen
C. Lens epithelium
D. Retinal ganglion cells
E. Cones

36. A 27-year-old woman has episodes that begin with a rising feeling in her stomach and intense déjà vu, followed by staring, chewing movements, and no memory of the event. Afterward she is confused for 20 minutes.

Where do these seizures most likely start?

A. Frontal lobe motor cortex
B. Occipital lobe
C. Cerebellum
D. Temporal lobe
E. Both hemispheres at once

37. A 77-year-old woman with DLB is given haloperidol in hospital for agitation. Within a day she becomes rigid, drowsy, and unable to swallow.

Which is the best explanation?

A. Natural progression of DLB over a few hours
B. A new stroke
C. A severe antipsychotic sensitivity reaction
D. Serotonin syndrome
E. Alcohol withdrawal

38. A lesion of the left internal capsule causes right-sided weakness, but a lesion of the left half of the cervical cord causes left-sided weakness.

Why?

A. Corticospinal fibers cross segment by segment in the anterior white commissure
B. The cord carries only the anterior corticospinal tract
C. Corticospinal fibers cross at the pyramidal decussation in the medulla
D. Corticospinal fibers cross in the posterior limb of the internal capsule
E. The rubrospinal tract crosses in the cord

39. Two months after starting a drug for mild Alzheimer disease, an 80-year-old man has a HR of 46/min and dizzy spells.

Which drug is most likely responsible?

A. Memantine
B. Lecanemab
C. Sertraline
D. Donepezil
E. Melatonin

40. Two patients have a first demyelinating attack. One has a conus lesion and positive MOG-IgG; the other has well-circumscribed periventricular lesions and CSF oligoclonal bands.

Which feature distinguishes the course of MOGAD from that of MS?

A. MOGAD is linked with other autoimmune diseases
B. MOGAD responds best to natalizumab
C. MOGAD is always monophasic
D. MOGAD has no progressive phase
E. MOGAD affects mainly women

41. A 58-year-old woman has a right eye that is down and out with a dilated pupil and ptosis, and weakness of her left arm and leg. She has no ataxia or tremor.

Which syndrome is this?

A. Benedikt syndrome
B. Claude syndrome
C. Foville syndrome
D. Millard-Gubler syndrome
E. Weber syndrome

42. A 64-year-old woman has weakness of the lower right face. She can wrinkle both sides of her forehead equally, and her right arm is weak.

Where is the lesion?

A. Right facial nucleus
B. Right facial nerve in the stylomastoid foramen
C. Left cortex or corticobulbar tract
D. Right cortex
E. Right internal acoustic meatus

43. A 27-year-old man has painful vision loss in both eyes over a week. Both optic discs are swollen, and MRI shows enhancement of most of each optic nerve, with perineural sheath enhancement.

Which antibody test is most likely to be positive?

A. MOG-IgG
B. Anti-acetylcholine receptor
C. Anti-NMDA receptor
D. AQP4-IgG
E. ANA

44. A 29-year-old woman taking the OCP has a week of worsening headache, then a seizure. CT shows a small hemorrhage near the cortex, and CT venography shows an empty delta sign.

Which treatment is most appropriate?

A. IV alteplase
B. Surgical evacuation
C. Anticoagulation
D. Avoid anticoagulation because of the hemorrhage
E. Embolization

45. After repair of a thoracic aortic aneurysm, a 69-year-old man cannot move his legs and has lost pain and temperature sensation below the umbilicus. Vibration and joint position sense in his toes are normal. He has urinary retention.

Which syndrome is this?

A. Brown-Séquard syndrome
B. Central cord syndrome
C. Anterior cord syndrome
D. Posterior cord syndrome
E. Medial medullary syndrome

46. A 74-year-old man with severe heart failure has cycles of breathing that grow deeper and then shallower, followed by pauses of 20 seconds, repeating about every minute during sleep.

Why does heart failure cause this pattern?

A. Upper airway collapse blocks airflow
B. Metabolic acidosis drives deep breathing through the chemoreceptors
C. Slow circulation delays the blood gas signal to the chemoreceptors
D. Pontine damage prolongs inspiration with pauses at full lung volume
E. Raised ICP compresses the medulla

47. A 30-year-old man with a complete C6 spinal cord injury has a sudden pounding headache, flushing above the injury, and a BP of 210/120 mm Hg with HR 48/min.

Which is the most appropriate first step?

A. Check and relieve bladder distension
B. Perform carotid sinus massage to slow the heart
C. Give IV furosemide to lower his blood volume
D. Give atropine
E. Lay him flat

48. A 58-year-old woman with open-angle glaucoma and asthma is wheezing more since starting a new eye drop.

Which drop is most likely responsible?

A. Pilocarpine
B. Dorzolamide
C. Latanoprost
D. Brimonidine
E. Timolol

49. A 44-year-old woman keeps bumping into door frames on both sides. Visual fields show loss of the outer half of the field in each eye.

Where is the lesion?

A. Temporal lobe through the Meyer loop
B. Left optic nerve
C. Right optic tract
D. Optic chiasm
E. Occipital cortex

50. A 33-year-old woman with MS notices that her leg weakness and blurred vision worsen after a hot shower and settle within an hour.

What is this phenomenon?

A. Lhermitte sign
B. An acute relapse
C. Charcot triad
D. Secondary progression
E. Uhthoff phenomenon

Test A answer sheet

Name _____ Date _____ Score _____ / 50

1 A B C D E	**14** A B C D E	**27** A B C D E	**40** A B C D E
2 A B C D E	**15** A B C D E	**28** A B C D E	**41** A B C D E
3 A B C D E	**16** A B C D E	**29** A B C D E	**42** A B C D E
4 A B C D E	**17** A B C D E	**30** A B C D E	**43** A B C D E
5 A B C D E	**18** A B C D E	**31** A B C D E	**44** A B C D E
6 A B C D E	**19** A B C D E	**32** A B C D E	**45** A B C D E
7 A B C D E	**20** A B C D E	**33** A B C D E	**46** A B C D E
8 A B C D E	**21** A B C D E	**34** A B C D E	**47** A B C D E
9 A B C D E	**22** A B C D E	**35** A B C D E	**48** A B C D E
10 A B C D E	**23** A B C D E	**36** A B C D E	**49** A B C D E
11 A B C D E	**24** A B C D E	**37** A B C D E	**50** A B C D E
12 A B C D E	**25** A B C D E	**38** A B C D E	
13 A B C D E	**26** A B C D E	**39** A B C D E	

Test B

50 questions · 60 minutes

1. A 28-year-old man is found unresponsive with pinpoint pupils and a respiratory rate of 4/min.

Where do opioids act to depress his breathing?

A. μ receptors at the preBötzinger complex
B. Peripheral chemoreceptors in the carotid bodies
C. The diaphragm
D. Lung stretch receptors
E. The pneumotaxic center in the upper pons

2. After surgery on the right temporal lobe for epilepsy, a man has lost the upper left quarter of his visual field in both eyes.

Which fibers were cut?

A. Right optic nerve
B. Meyer loop
C. Left optic tract
D. Optic chiasm
E. Parietal optic radiations

3. A 32-year-old woman with NMOSD is planning pregnancy and asks how pregnancy will affect her disease.

Which statement is correct?

A. Pregnancy has no effect on relapse risk
B. Pregnancy cures NMOSD in most women
C. Relapses increase during pregnancy and after delivery
D. Relapses usually fall during pregnancy, as in MS
E. Pregnancy usually turns NMOSD into a progressive disease

4. A man with a cord lesion has a sensory level at the umbilicus.

Which dermatome is this?

A. L1
B. L4
C. T4
D. T10
E. T6

5. The wife of a 59-year-old man with bvFTD asks what can help his disinhibition and compulsive behavior.

Which drug is most appropriate?

A. Donepezil
B. An SSRI
C. Memantine
D. High-dose haloperidol
E. Levodopa

6. A 48-year-old factory worker has tinnitus and difficulty hearing conversation. Audiometry shows a dip at 4 kHz with recovery at 8 kHz in both ears.

What is the cause?

A. Aminoglycosides
B. Otosclerosis
C. Presbycusis
D. Ménière disease
E. Noise

7. A dentist injects lidocaine before filling a tooth, and the patient feels no pain.

What is the mechanism?

A. Inhibits the Na/K-ATPase
B. Activates GABA-A receptors
C. Blocks voltage-gated sodium channels in the nerve
D. Opens potassium leak channels
E. Blocks voltage-gated calcium channels at the terminal

8. A 9-year-old boy has encephalopathy and multifocal deficits 2 weeks after a viral illness. MRI shows fluffy, ill-defined white matter lesions, and serum MOG-IgG is positive at 1:320.

Where is the target antigen of this disease?

A. Peripheral nerve myelin made by Schwann cells
B. Water channels on astrocyte end-feet
C. The outermost layer of myelin and the oligodendrocyte surface
D. Voltage-gated calcium channels
E. Nicotinic acetylcholine receptors at the neuromuscular junction

9. A student traces the direct pathway of the basal ganglia.

What is its net effect on movement?

A. The GPi excites the thalamus to drive movement
B. The STN inhibits the thalamus directly
C. Dopamine inhibits D1 neurons to reduce movement
D. D1 neurons inhibit the GPi, releasing the thalamus and allowing movement
E. D2 neurons inhibit the GPe, which frees the STN to excite the GPi and restrain movement

10. A 35-year-old man who often sits with his legs crossed has a right foot drop. Dorsiflexion and eversion are weak, the dorsum of the foot is numb, but inversion is strong.

Which nerve is injured?

A. Femoral nerve
B. Common peroneal nerve at the fibular head
C. Sciatic nerve
D. Tibial nerve
E. L5 root

11. A 34-year-old woman has a stroke the day after a long-haul flight, during which her calf became swollen. Workup finds no A-fib or carotid disease, and a bubble echocardiogram shows a patent foramen ovale.

Which mechanism best explains her stroke?

A. Global hypoperfusion causing a watershed infarct in the cortex
B. A venous thrombus crossing a right-to-left shunt
C. Lipohyalinosis of penetrating arteries
D. A left atrial thrombus from A-fib
E. Atherosclerotic thrombus in the ICA

12. Several young adults who injected a synthetic opioid made in a home laboratory developed sudden, severe parkinsonism.

How did the contaminant damage dopamine neurons?

A. It blocked D2 receptors in the striatum
B. It was taken up by DAT and impaired mitochondria
C. It depleted vesicular dopamine by blocking VMAT2
D. It caused autoantibodies against the SNc
E. It expanded a CAG repeat

13. A 30-year-old woman whose father has Huntington disease has predictive testing. She has 38 CAG repeats.

How is this result interpreted?

A. Reduced penetrance range
B. Full penetrance with juvenile onset
C. Intermediate allele that never causes disease
D. Normal
E. Diagnostic of manifest disease

14. A 67-year-old man arrives 50 minutes after sudden right face and arm weakness and difficulty speaking. His BP is 176/98 mm Hg.

Which of the following should be done first?

A. MRI with diffusion-weighted imaging
B. Aspirin 300 mg
C. Glucose and non-contrast CT head
D. Lower his BP to below 140/90
E. IV alteplase

15. A 67-year-old woman has brief spinning lasting under a minute when she rolls over in bed or looks up. Hearing is normal. The Dix-Hallpike maneuver to the right provokes upbeat, torsional nystagmus that fatigues with repetition.

Which is the most appropriate treatment?

A. MRI and MRA
B. The Epley maneuver
C. Intratympanic gentamicin
D. Meclizine daily for 3 months
E. Oral prednisone

16. An 82-year-old man has a supratentorial hemorrhage estimated at 40 mL on CT, with intraventricular extension. His GCS is 9.

What is his ICH score?

A. 2
B. 5
C. 4
D. 6
E. 3

17. A 63-year-old man has mild memory problems, and the diagnosis is uncertain after clinical assessment and MRI.

Which CSF pattern supports Alzheimer disease?

A. Increased Aβ42 with decreased tau and phosphorylated tau
B. Oligoclonal bands with raised white cells
C. Decreased Aβ42 with increased tau and phosphorylated tau
D. Raised 14-3-3 protein only
E. Normal Aβ42 and normal tau

18. A 42-year-old man with Down syndrome develops progressive memory loss and loss of daily skills.

Why is Alzheimer disease more common and earlier in Down syndrome?

A. Loss of APOE on chromosome 21
B. Reduced cholinergic input to the cortex from birth
C. An extra copy of the tau gene
D. An extra copy of the APP gene on chromosome 21
E. Mutations in PSEN2

19. A 49-year-old woman had a ruptured aneurysm coiled 6 days ago. Today she becomes confused and develops new right arm weakness. CT head shows no new bleeding.

Which complication is most likely?

A. Vasospasm and delayed cerebral ischemia
B. Hemorrhagic transformation
C. Central post-stroke pain
D. Malignant MCA infarct from the initial bleed
E. Rebleeding

20. A healthy volunteer is given an infusion that raises his BP, and his HR falls.

Which pathway lowers sympathetic output?

A. Baroreceptors inhibit the NTS
B. RVLM excites the CVLM, which then inhibits the spinal cord
C. NTS inhibits the nucleus ambiguus and its cardiac vagal output
D. NTS excites the CVLM, which inhibits the RVLM
E. The area postrema inhibits the heart directly

21. A neurotransmitter binds a postsynaptic receptor and makes the membrane potential more negative.

Which receptor and ion best explain this?

A. AMPA receptor, sodium entry
B. NMDA receptor, calcium entry
C. Voltage-gated calcium channel, calcium entry
D. Nicotinic receptor, sodium entry
E. GABA-A receptor, chloride entry

22. A 74-year-old man says he can read the newspaper without his glasses for the first time in years, but his distance vision is worse and colors look yellowed.

Which type of cataract explains this?

A. Posterior subcapsular cataract
B. Cortical cataract
C. Snowflake cataract
D. Congenital cataract
E. Nuclear sclerosis

23. A 62-year-old man has a left putaminal hemorrhage on CT. His BP is 196/110 mm Hg, and he takes no anticoagulants.

Which is the most appropriate BP management?

A. Lower to below 185/110 before thrombolysis
B. Oral amlodipine started tomorrow
C. Allow BP up to 220/120 to preserve perfusion around the bleed
D. Nimodipine
E. IV nicardipine to a systolic target of 140

24. A 38-year-old man with flank pain and hematuria has enlarged kidneys full of cysts on ultrasound. His mother died suddenly of a brain hemorrhage at 45.

Which type of brain hemorrhage is he at increased risk of?

A. Subarachnoid hemorrhage from a saccular aneurysm
B. Venous infarction from sinus thrombosis
C. Subdural hemorrhage from bridging veins
D. Deep hemorrhage from Charcot-Bouchard aneurysms in the pons
E. Lobar hemorrhage from amyloid angiopathy

25. A woman with postherpetic neuralgia starts pregabalin.

What is its target?

A. The α2δ-1 subunit of voltage-gated calcium channels
B. μ-opioid receptors
C. COX-2
D. GABA-A receptors
E. NMDA receptors

26. A 24-year-old woman who sleeps in her contact lenses has 2 days of a painful red right eye, photophobia, and blurred vision. Slit lamp examination shows a white stromal infiltrate with an overlying epithelial defect.

Which is the most appropriate next step?

A. Topical corticosteroids
B. Oral acetazolamide
C. Artificial tears and cool compresses, with review in a week
D. Topical antihistamine drops
E. Corneal scraping, then urgent topical antibiotics

27. A neurologist orders DAT SPECT for a man with parkinsonism, and uptake is reduced.

Which diagnosis can this result exclude?

A. Corticobasal degeneration
B. Typical Parkinson disease
C. Multiple system atrophy
D. Drug-induced parkinsonism
E. Progressive supranuclear palsy

28. A 52-year-old man with high myopia has new floaters and flashing lights in his right eye, and now a dark shadow spreading from the side of his vision like a curtain.

Which is the most appropriate management?

A. IV methylprednisolone
B. AREDS supplements
C. Intravitreal anti-VEGF injection
D. Reassurance and review in 6 weeks
E. Urgent surgical repair

29. A 29-year-old woman has double vision. On looking left, her right eye fails to adduct while her left eye shows nystagmus, and the same happens on looking right. Convergence is normal.

Damage to which structure explains this?

A. Paramedian pontine reticular formation
B. Optic chiasm
C. Medial longitudinal fasciculus
D. Bilateral abducens nerves
E. Bilateral oculomotor nerves

30. Two days after SAH, a 60-year-old man becomes increasingly drowsy. CT shows enlarged lateral ventricles.

What is the mechanism of this complication?

A. Blood blocks CSF absorption at the arachnoid granulations
B. The aneurysm compresses the cerebral aqueduct from the start
C. Hyponatremia causes cerebral edema
D. Vasospasm infarcts the periventricular white matter
E. Blood stimulates the choroid plexus to make more CSF

31. A comatose patient with a pontine hemorrhage extends all four limbs to pain.

Which tracts drive this posture?

A. The medullary reticulospinal tract acting on extensors
B. The lateral corticospinal tract
C. Unopposed vestibulospinal and pontine reticulospinal tracts
D. The rubrospinal tract from an intact red nucleus
E. The dorsal column-medial lemniscus

32. A 23-year-old woman has a routine 30-minute EEG between her episodes of convulsive collapse, and it is normal.

How should this result be interpreted?

A. It confirms syncope
B. It does not exclude epilepsy
C. It excludes focal epilepsy but not generalized epilepsy
D. It confirms PNES
E. It means no further testing is needed

33. A 45-year-old man has had 3 attacks of vertigo lasting 2–4 hours, each with roaring tinnitus and fullness in his left ear. Audiometry shows left low-frequency sensorineural loss.

Which is the most appropriate first treatment?

A. Surgical section of the vestibular nerve as first-line treatment
B. Intratympanic gentamicin
C. The Epley maneuver
D. Less salt, caffeine, and alcohol, with a thiazide or betahistine
E. Long-term daily meclizine

34. A newborn of hearing parents fails the hearing screen and has severe bilateral SNHL with no other abnormalities. Congenital CMV testing is negative.

Mutations in which gene are most likely?

A. EDNRB
B. PAX3
C. GJB2
D. MITF
E. The merlin gene

35. A neurologist notes that her patient with DLB draws a clock poorly and loses track of conversations, yet recalls recent events well.

Which feature of DLB explains the relatively spared memory?

A. Lewy bodies in the occipital cortex
B. Lewy bodies in the dorsal motor nucleus of the vagus
C. Dopamine loss from the substantia nigra
D. Loss of the nucleus basalis of Meynert
E. Relative preservation of the medial temporal lobe

36. A 62-year-old East Asian woman has sudden severe pain in her right eye, halos around lights, headache, and vomiting after an evening in a dark cinema. The eye is red and firm, the cornea hazy, and the pupil fixed and mid-dilated. IOP is 58 mmHg.

After lowering the IOP medically, which treatment is definitive?

A. Topical latanoprost long term
B. Topical corticosteroids
C. Topical atropine
D. Laser peripheral iridotomy
E. Laser trabeculoplasty

37. A 26-year-old woman with obesity has daily headaches, pulsatile tinnitus, and horizontal double vision. Both optic discs are swollen, and MRI shows no mass.

Which is the next step?

A. EEG
B. Lumbar puncture with opening pressure
C. Reassurance; this is tension-type headache
D. 100% oxygen
E. Temporal artery biopsy

38. When a patient says "ah," the uvula deviates to the right.

Which nerve is damaged?

A. Right CN IX
B. Right CN X
C. Left CN XII
D. Left CN XI
E. Left CN X

39. A 45-year-old man with epilepsy is found after a witnessed tonic-clonic seizure with a bitten tongue. An hour later he has left arm weakness that is improving. CT head is normal and glucose is normal.

Which is the most appropriate management of his weakness?

A. IV methylprednisolone
B. Start aspirin and clopidogrel
C. CT angiogram and mechanical thrombectomy
D. Observation, expecting it to resolve
E. IV alteplase

40. A 70-year-old man with a COPD exacerbation is given high-flow oxygen, and his $PaCO_2$ rises.

Which explanation is most accepted?

A. Oxygen paralyzes the diaphragm
B. Loss of his hypoxic drive makes him stop breathing for long periods
C. Oxygen blocks the central chemoreceptors on the ventral medulla
D. Oxygen stimulates the apneustic center
E. Ventilation-perfusion mismatch and the Haldane effect

41. In Parkinson disease, SNc dopamine neurons die.

What happens to GPi output?

A. It is unchanged
B. It becomes excitatory
C. It stops entirely, causing ballismus
D. It decreases, releasing the VA and VL thalamus and causing chorea
E. It increases, inhibiting the thalamus and causing bradykinesia

42. A 70-year-old man has right facial weakness including the forehead, ear pain, and vesicles in his right ear canal.

What is the cause?

A. Varicella zoster of the facial nerve
B. Vestibular schwannoma
C. Bell palsy
D. Trigeminal neuralgia
E. A left cortical stroke

43. A woman with idiopathic intracranial hypertension has horizontal double vision and cannot abduct her left eye.

Why is CN VI a false localizing sign?

A. It crosses before leaving the brainstem
B. Its long course up the clivus makes it vulnerable to raised ICP
C. Its parasympathetic pupil fibers lie on the surface and are compressed first
D. It is the only cranial nerve to leave the back of the brainstem
E. It passes through the parotid gland

44. An 8-year-old girl's teacher reports many brief staring spells a day, with lip smacking, after which she immediately resumes her work. Hyperventilation in clinic triggers one, and EEG shows a 3 Hz spike-and-wave pattern.

Which drug is most appropriate?

A. Ethosuximide
B. Vigabatrin
C. Carbamazepine
D. Phenytoin
E. Lacosamide

45. A 52-year-old woman has a tremor of both hands when holding a cup or writing, which improves with a glass of wine. Her father had the same tremor. There is no bradykinesia.

Which drug is most appropriate?

A. Trihexyphenidyl
B. Haloperidol
C. Propranolol
D. Levodopa
E. Tetrabenazine

46. A 55-year-old man on dialysis needs imaging for a suspected brain abscess.

Why is gadolinium avoided where possible?

A. It makes blood bright on SWI
B. Risk of NSF
C. It cannot show enhancing lesions
D. It is radioactive
E. Risk of contrast nephropathy from iodine

47. A toxin prevents calcium from entering the presynaptic terminal when an action potential arrives.

Which step of transmission fails?

A. Opening of postsynaptic chloride channels
B. Enzymatic breakdown of acetylcholine
C. Reuptake of transmitter
D. SNARE-mediated vesicle fusion
E. Propagation along the axon

48. A 61-year-old woman with MSA faints when she stands up. Her BP falls from 132/80 lying to 88/56 mm Hg standing.

Which drug is most appropriate?

A. Amlodipine
B. Furosemide
C. Botulinum toxin
D. Midodrine
E. Tamsulosin

49. A 42-year-old man has restless, flowing, involuntary movements of his fingers, face, and trunk, with irritability and poor planning.

Loss of which neurons first explains his movements?

A. Indirect-pathway medium spiny neurons projecting to the GPe

B. Direct-pathway D1 medium spiny neurons projecting to the GPi

C. Dopamine neurons of the SNc

D. Cholinergic neurons of the nucleus basalis

E. Purkinje cells of the cerebellum

50. A 56-year-old man has 18 months of making crude jokes to strangers, shoplifting, eating whole packets of biscuits, and showing no concern when his wife was ill. He denies anything has changed. Memory and drawing are intact.

Which diagnosis is most likely?

A. Behavioral variant frontotemporal dementia

B. Depression

C. Alzheimer disease

D. Dementia with Lewy bodies

E. Huntington disease

Test B answer sheet

Name _____ Date _____ Score _____ / 50

1 A B C D E	**14** A B C D E	**27** A B C D E	**40** A B C D E
2 A B C D E	**15** A B C D E	**28** A B C D E	**41** A B C D E
3 A B C D E	**16** A B C D E	**29** A B C D E	**42** A B C D E
4 A B C D E	**17** A B C D E	**30** A B C D E	**43** A B C D E
5 A B C D E	**18** A B C D E	**31** A B C D E	**44** A B C D E
6 A B C D E	**19** A B C D E	**32** A B C D E	**45** A B C D E
7 A B C D E	**20** A B C D E	**33** A B C D E	**46** A B C D E
8 A B C D E	**21** A B C D E	**34** A B C D E	**47** A B C D E
9 A B C D E	**22** A B C D E	**35** A B C D E	**48** A B C D E
10 A B C D E	**23** A B C D E	**36** A B C D E	**49** A B C D E
11 A B C D E	**24** A B C D E	**37** A B C D E	**50** A B C D E
12 A B C D E	**25** A B C D E	**38** A B C D E	
13 A B C D E	**26** A B C D E	**39** A B C D E	

Test C

50 questions · 60 minutes

1. A 38-year-old man who carries 44 CAG repeats has no symptoms yet.

Which imaging finding may already be present?

A. Caudate atrophy
B. Asymmetric frontoparietal atrophy
C. Midbrain atrophy
D. Medial temporal atrophy
E. Hot cross bun sign

2. A 28-year-old woman has optic neuritis. MRI shows T2 plaques in periventricular and juxtacortical sites, all non-enhancing. She has had no previous symptoms.

Under the 2024 McDonald criteria, which result would most directly confirm MS now?

A. CSF oligoclonal bands
B. Positive AQP4 antibody
C. A raised ESR
D. Visual evoked potentials showing slowed conduction
E. A normal B12 level

3. Dopamine from the SNc acts on both striatal output pathways.

How does it favor movement through both?

A. It excites both D1 and D2 neurons equally in the striatum
B. It excites D1 neurons and inhibits D2 neurons
C. It excites the STN directly
D. It inhibits D1 neurons and excites D2 neurons, braking movement
E. It inhibits the thalamus

4. A 73-year-old man has a year of fluctuating confusion, with good and bad days, and sees detailed small children in his living room. His wife says he has acted out his dreams for 10 years. He has mild bradykinesia.

Which diagnosis is most likely?

A. Dementia with Lewy bodies
B. Delirium
C. Frontotemporal dementia
D. Alzheimer disease
E. Parkinson disease dementia

5. A 30-year-old woman has drooping eyelids and double vision that worsen through the day, with fatigable arm weakness.

Which is the first-line symptomatic drug?

A. Sertraline
B. Pyridostigmine
C. Lorazepam
D. Atropine
E. Botulinum toxin

6. A 55-year-old woman has raised IOP of 28 mmHg in both eyes on routine screening.

Which examination determines whether she has open-angle or angle-closure glaucoma?

A. Fundoscopy
B. Visual field testing
C. Tonometry
D. Gonioscopy
E. OCT of the nerve fiber layer

7. A man's tongue deviates to the left on protrusion, with wasting and fasciculations of the left side.

Which nerve is damaged?

A. Left CN IX
B. Left CN XII
C. Right corticobulbar tract
D. Left CN X
E. Right CN XII

8. A 38-year-old woman has sudden weakness of her whole right face, with sounds seeming too loud and loss of taste on the front of the tongue.

Which branch explains the hyperacusis?

A. Posterior auricular nerve
B. V3
C. Greater petrosal nerve
D. Chorda tympani
E. Nerve to stapedius

9. A 70-year-old man notices his eyelashes have grown longer and his light brown iris has darkened since starting a glaucoma drop.

How does this drop lower IOP?

A. Constricts ciliary body vessels through α1 receptors
B. Reduces bicarbonate in the ciliary body
C. Contracts the ciliary muscle to open the trabecular meshwork
D. Decreases aqueous production by blocking ciliary β receptors
E. Increases uveoscleral outflow

10. A 30-year-old woman has a first attack of myelitis. CSF shows 85 white cells/μL with neutrophils and eosinophils and no oligoclonal bands.

Which diagnosis do these CSF findings favor?

A. Small vessel disease
B. Functional neurologic disorder
C. Multiple sclerosis
D. NMOSD
E. B12 deficiency

11. A 60-year-old man has small, irregular pupils that constrict when he looks at a near target but not to light.

Which condition do these pupils mark?

A. Neurosyphilis
B. CN III palsy
C. Opioid toxicity
D. Optic neuritis
E. Horner syndrome

12. A 14-year-old girl has progressive clumsiness. Examination shows ataxia of the right arm and leg.

If a spinocerebellar tract is responsible, on which side is the lesion?

A. It cannot be determined
B. The midline only
C. Right
D. Left
E. Both sides equally

13. A 76-year-old woman has 2 years of progressive short-term memory loss and now gets lost driving. MMSE is 21/30. TFTs, B12, and electrolytes are normal, and MRI shows medial temporal atrophy.

Which drug is most appropriate to start?

A. Donepezil
B. Haloperidol
C. Memantine alone
D. Tetrabenazine
E. Levodopa

14. A neuroanatomy student traces vibration sense from the right foot to the cortex.

Where do these fibers cross the midline?

A. They never cross
B. In the anterior white commissure of the cord
C. In the pons
D. In the thalamus
E. In the medulla, as internal arcuate fibers

15. A man with an expanding intracerebral hemorrhage develops a BP of 220/90 mm Hg, HR 44/min, and irregular breathing.

Why does his BP rise?

A. A sympathetic surge to maintain CPP as ICP rises
B. Baroreceptor failure from damage to the nucleus tractus solitarius
C. Pain from the headache
D. Release of ANP from the atria
E. Activation of the vagus nerve by the expanding hematoma

16. A 70-year-old man with hypertension has equal weakness of his right face, arm, and leg, with no aphasia, neglect, or visual field loss.

Where is the lesion most likely?

A. Left ACA territory
B. Right pons
C. Left cervical cord
D. Left MCA cortex on the lateral convexity
E. Left internal capsule

17. A 41-year-old man woke with a droop of the right side of his face. He cannot wrinkle his right forehead or close his right eye fully. Limb strength, speech, and sensation are normal.

Why is this unlikely to be a stroke?

A. Facial weakness from stroke is always bilateral
B. Facial weakness that includes the forehead points to a peripheral lesion
C. Strokes never affect the face
D. Strokes affecting the face always cause arm weakness as well
E. He has no vascular risk factors

18. A 61-year-old man has a left eye that cannot abduct and weakness of his right arm and leg.

Where is the lesion?

A. Right pons
B. Left pons
C. Left internal capsule
D. Left cervical cord
E. Left midbrain

19. MRI in a woman with acute paraplegia shows a swollen, central cord lesion spanning 4 vertebral segments with enhancement. Brain MRI shows a lesion in the dorsal medulla.

Which diagnosis does this pattern most suggest?

A. Lacunar stroke
B. Multiple sclerosis
C. Posterior cord syndrome from B12 deficiency
D. NMOSD
E. Central cord syndrome from hyperextension

20. A 19-year-old woman has an episode of thrashing with her eyes closed, side-to-side head movement, and asynchronous limb movements for 4 minutes. She is alert immediately afterward.

Which test best confirms the diagnosis?

A. Lumbar puncture
B. Toxicology screen
C. Video EEG
D. Serum prolactin 10–20 minutes later
E. MRI brain

21. A 72-year-old man is booked for cataract surgery. His medications include tamsulosin for prostate symptoms.

Why should the surgeon know about tamsulosin?

A. It raises IOP
B. It is a contraindication to an IOL
C. It causes posterior subcapsular cataract
D. It increases bleeding risk
E. It causes a floppy iris during surgery

22. A 61-year-old man with an MI has pain in his left neck, shoulder, and back.

What explains why the pain is felt there?

A. C fibers from the heart project directly to the arm area of S1
B. Descending pathways redirect the pain
C. Visceral and somatic afferents converge on the same spinal segments
D. The phrenic nerve carries cardiac pain
E. The heart is innervated by branches of the brachial plexus

23. A 30-year-old man of Han Chinese ancestry with focal seizures is about to start carbamazepine.

Which test should be done first?

A. Thiamine level
B. HLA-B*1502
C. TPMT activity
D. G6PD level
E. HLA-B27

24. A 21-year-old woman collapsed after standing for an hour at a hot concert. She felt lightheaded and her vision greyed first, had a few brief jerks, and was fully alert within a minute.

Which tests are most appropriate?

A. Video EEG
B. MRI brain with gadolinium
C. ECG and orthostatic BP
D. DAT SPECT
E. Lumbar puncture

25. A 71-year-old man with DLB punches and kicks in his sleep while acting out dreams, and his wife has been hurt.

Which treatment is most appropriate first?

A. Melatonin
B. Quetiapine
C. Haloperidol
D. Pramipexole
E. Amitriptyline

26. A student asks which corticospinal fibers remain uncrossed until their segment.

What do these fibers serve?

A. Facial expression
B. Axial and proximal muscles on both sides
C. Skilled distal movements of the opposite hand and fingers
D. Upper limb flexion
E. Eye movements

27. A 50-year-old woman with SAH is awaiting coiling of her aneurysm later today and has a severe headache.

Which analgesic is most appropriate?

A. Paracetamol
B. Ketorolac
C. Aspirin
D. Diclofenac
E. Ibuprofen

28. A 52-year-old man with behavioral change and frontal atrophy develops fasciculations and wasting of his hands. His father had ALS.

Which genetic cause is most likely?

A. MAPT mutation
B. HTT CAG expansion
C. PSEN1 mutation
D. C9orf72 repeat expansion
E. APOE ε4

29. A 30-year-old man with MOGAD has a second attack of optic neuritis a year after his first, and MOG-IgG remains positive.

Which treatment is most appropriate to prevent further relapses?

A. IVIG every 4 weeks
B. No treatment, since MOGAD is monophasic
C. Glatiramer acetate
D. Fingolimod
E. Interferon beta

30. A 52-year-old woman with a BMI of 48 has daytime sleepiness and a chronically raised $PaCO_2$ with a high bicarbonate.

Why does her hypercapnia persist?

A. She has metabolic acidosis
B. Her lung stretch receptors fire continuously and inhibit inspiration
C. Retained bicarbonate blunts the central chemoreceptors
D. Her preBötzinger complex is absent
E. Her carotid bodies have been destroyed by chronic hypoxia

31. A student measures the resting membrane potential of a neuron at about −70 mV.

Why is it close to the potassium equilibrium potential?

A. Calcium entry hyperpolarizes the membrane
B. The Na/K-ATPase pumps more potassium out of the cell than sodium in
C. Chloride channels are closed
D. Potassium leak channels make the membrane most permeable to potassium at rest
E. Voltage-gated sodium channels stay partly open throughout rest

32. A 70-year-old right-handed woman suddenly cannot read her newspaper, although she can still write a letter. She has a right homonymous hemianopia with macular sparing and no weakness.

Which artery is most likely occluded?

A. Left anterior cerebral artery
B. Left middle cerebral artery
C. Left ophthalmic artery
D. Left posterior cerebral artery
E. Right posterior cerebral artery

33. A 78-year-old smoker notices that straight lines look wavy in her left eye over a week. An Amsler grid shows a central scotoma, and OCT shows subretinal fluid.

What drives the new vessels causing her symptoms?

A. Rod photoreceptor death
B. Lipofuscin
C. VEGF
D. Drusen
E. Emboli from the carotid artery

34. An 81-year-old man faints when he turns his head while shaving. Monitored carotid sinus massage reproduces his symptoms with 5 seconds of asystole.

Which treatment is most appropriate?

A. Fludrocortisone
B. Tilt training
C. Midodrine
D. Reassurance only
E. Permanent pacemaker

35. A 79-year-old woman has a lobar hemorrhage and no hypertension.

Which test best supports cerebral amyloid angiopathy?

A. Carotid duplex
B. DAT SPECT
C. EEG
D. MRI with SWI
E. Repeat non-contrast CT

36. A 4-year-old girl has hearing loss, a white forelock, one blue and one brown eye, and widely spaced inner canthi. Her father has a white forelock.

What is the underlying defect?

A. Endolymphatic hydrops
B. Loss of connexin gap junctions in the cochlea
C. Failed migration of neural crest cells
D. Ototoxic drug exposure in utero
E. Congenital CMV infection

37. A 38-year-old woman has 2 days of continuous severe vertigo and vomiting after a cold. Hearing is normal. She has unidirectional horizontal-torsional nystagmus suppressed by fixation, an abnormal head impulse to the right, and no skew.

What is the most likely diagnosis?

A. Posterior circulation stroke
B. Ménière disease
C. BPPV
D. Labyrinthitis
E. Vestibular neuritis

38. A 47-year-old man has an aneurysmal SAH that is coiled within 24 hours of onset.

Which drug should he receive to prevent delayed ischemia?

A. Levetiracetam
B. Mannitol
C. Aspirin
D. Tranexamic acid for 2 weeks
E. Nimodipine

39. A student notes that smell differs from the other senses in its route to the cortex.

What is distinctive about it?

A. It crosses entirely at the chiasm
B. Its receptor neurons never regenerate once they are lost
C. It reaches cortex without a thalamic relay
D. It is carried by the trigeminal nerve
E. It relays in the medial geniculate nucleus before the cortex

40. A comatose man extends and internally rotates all four limbs in response to a painful stimulus.

Where is the lesion most likely?

A. In the cerebellar vermis
B. In the upper cervical spinal cord
C. Above the red nucleus
D. At or below the red nucleus, in the midbrain or rostral pons
E. In the peripheral nerves

41. A 74-year-old man suddenly develops violent, flinging movements of his left arm and leg.

Where is the lesion most likely?

A. Right cerebellar hemisphere
B. Left caudate
C. Right subthalamic nucleus
D. Right substantia nigra
E. Left subthalamic nucleus

42. A 31-year-old woman with relapsing-remitting MS has 5 days of new numbness and weakness of her left leg. Infection has been excluded.

Which is the most appropriate treatment?

A. Plasma exchange first
B. IV methylprednisolone for 3–7 days
C. IVIG
D. Start natalizumab today
E. Baclofen

43. A 54-year-old man has slowly worsening hearing in his right ear with ringing on the same side and mild imbalance. Audiometry shows right sensorineural loss.

Which investigation is most appropriate?

A. GJB2 testing
B. High-resolution CT of the temporal bones without contrast
C. Tympanometry only
D. Reassurance; this is presbycusis
E. MRI with gadolinium of the internal auditory canals

44. A 68-year-old man has a bilateral tremor of his hands when holding a cup. His neurologist is unsure whether this is Parkinson disease.

Which result would point away from Parkinson disease?

A. Reduced putaminal uptake on DAT SPECT
B. Normal striatal uptake on DAT SPECT
C. Levodopa-induced dyskinesia
D. Olfactory loss
E. A dramatic response to levodopa

45. A 26-year-old woman cannot adduct her right eye on looking left, while her left eye shows nystagmus. Convergence is normal.

Where is the lesion?

A. Left MLF
B. Right MLF
C. Right CN III
D. Left CN VI nucleus
E. Right paramedian pontine reticular formation

46. A 71-year-old woman's family notices she has become apathetic and is wetting herself. Examination shows weakness and reduced sensation of the right leg, much worse than the right arm, and a grasp reflex.

Which artery is most likely affected?

A. Left anterior cerebral artery
B. Right anterior cerebral artery
C. Left middle cerebral artery
D. Left posterior cerebral artery
E. Left lenticulostriate arteries

47. A 78-year-old woman with progressive memory loss has a headache and left homonymous hemianopia. CT shows a right occipital lobar hemorrhage. MRI with SWI shows many lobar microbleeds. She has no history of hypertension.

What is the most likely cause?

A. Cerebral venous sinus thrombosis
B. Arteriovenous malformation
C. Hypertension
D. Mycotic aneurysm
E. Cerebral amyloid angiopathy

48. Non-contrast CT 40 minutes after sudden left hemiplegia shows a bright, linear density in the right MCA but no hemorrhage.

What does this sign indicate?

A. An old infarct
B. Normal arterial flow
C. Subarachnoid blood
D. Thrombus in the artery, highly specific for LVO
E. A calcified aneurysm of the MCA

49. Six months after pramipexole was added for Parkinson disease, a 62-year-old man has spent his savings on online gambling.

Which is the most appropriate management?

A. Add rasagiline
B. Increase levodopa and continue pramipexole
C. Reduce or stop pramipexole
D. Refer for deep brain stimulation
E. Start an SSRI and continue pramipexole

50. A 58-year-old man has 2 years of parkinsonism with little benefit from levodopa, fainting on standing, urinary incontinence, erectile dysfunction, and a wide-based unsteady gait. His wife reports loud snoring with pauses and noisy breathing.

Which MRI finding is most likely?

A. Hummingbird sign of the midbrain
B. Medial temporal atrophy
C. Asymmetric frontoparietal atrophy
D. Hot cross bun sign in the pons
E. Caudate atrophy

Test C answer sheet

Name _____ Date _____ Score _____ / 50

1 A B C D E	**14** A B C D E	**27** A B C D E	**40** A B C D E
2 A B C D E	**15** A B C D E	**28** A B C D E	**41** A B C D E
3 A B C D E	**16** A B C D E	**29** A B C D E	**42** A B C D E
4 A B C D E	**17** A B C D E	**30** A B C D E	**43** A B C D E
5 A B C D E	**18** A B C D E	**31** A B C D E	**44** A B C D E
6 A B C D E	**19** A B C D E	**32** A B C D E	**45** A B C D E
7 A B C D E	**20** A B C D E	**33** A B C D E	**46** A B C D E
8 A B C D E	**21** A B C D E	**34** A B C D E	**47** A B C D E
9 A B C D E	**22** A B C D E	**35** A B C D E	**48** A B C D E
10 A B C D E	**23** A B C D E	**36** A B C D E	**49** A B C D E
11 A B C D E	**24** A B C D E	**37** A B C D E	**50** A B C D E
12 A B C D E	**25** A B C D E	**38** A B C D E	
13 A B C D E	**26** A B C D E	**39** A B C D E	

Test A answers and explanations

1 **D**	2 **B**	3 **B**	4 **D**	5 **A**	6 **A**	7 **C**	8 **A**	9 **B**	10 **C**
11 **D**	12 **A**	13 **E**	14 **E**	15 **A**	16 **B**	17 **A**	18 **B**	19 **E**	20 **B**
21 **E**	22 **D**	23 **C**	24 **B**	25 **C**	26 **D**	27 **A**	28 **B**	29 **D**	30 **B**
31 **E**	32 **E**	33 **B**	34 **E**	35 **A**	36 **D**	37 **C**	38 **C**	39 **D**	40 **D**
41 **E**	42 **C**	43 **A**	44 **C**	45 **C**	46 **C**	47 **A**	48 **E**	49 **D**	50 **E**

1. D. Delirium

Alzheimer disease · Step 2

Delirium is favored by sudden onset, fluctuating attention, and an acute illness, and prompts a search for the cause.

2. B. Astrocyte

NMOSD · Step 1

AQP4-IgG binds AQP4 on astrocyte end-feet and kills astrocytes through the classical complement pathway, so demyelination is secondary, with oligodendrocytes and axons damaged as bystanders.

C. Oligodendrocytes are the target in MOGAD and are damaged only secondarily in NMOSD.

3. B. Phacomorphic glaucoma; lens removal

Cataract · Step 1

In phacomorphic glaucoma, a swollen lens pushes the iris forward and closes the angle. In phacolytic glaucoma, a hypermature lens leaks proteins and macrophages clog the trabecular meshwork while the angle stays open. Both are treated by removing the lens.

E. Phacolytic glaucoma has an open angle.

4. D. Bilateral corticobulbar lesions

Physiology · Descending tracts · Step 1

Bilateral corticobulbar lesions cause pseudobulbar palsy, with dysarthria, dysphagia, and a brisk jaw jerk, without the wasting of an LMN lesion.

5. A. Unmyelinated C fibers

Physiology · Pain pathways and descending modulation · Step 1

Lightly myelinated Aδ fibers carry the fast, sharp, well-localized first pain, and unmyelinated C fibers, with large receptive fields, carry slow, poorly localized pain that conveys intensity.

6. A. Lumbar puncture for xanthochromia

Subarachnoid hemorrhage · Step 2

CT within 6 hours of onset has a sensitivity near 100%, but after 6 hours sensitivity falls, so a lumbar puncture looking for xanthochromia follows a normal scan when suspicion remains.

D. Her CT was done after 6 hours, so it cannot rule out SAH alone.

7. C. Corticobasal degeneration

Atypical parkinsonism · Step 2

CBD causes asymmetric limb rigidity, dystonia, and myoclonus, with limb apraxia, cortical sensory loss, and alien limb. MRI shows asymmetric frontoparietal atrophy.

8. A. Artificial tears, cool compresses, and hand hygiene

Vision loss and red eye · Step 2

Viral conjunctivitis, from adenovirus in up to 90%, causes a red eye with watery discharge and normal vision, often with follicles and preauricular nodes after a respiratory infection. It resolves within 3 weeks, and antibiotics do not help.

D. Urgent Gram stain is for hyperacute gonococcal conjunctivitis with copious purulent discharge.

9. B. Hearing becomes bilateral above the cochlear nuclei

Physiology · I, II, and VIII · Step 1

Hearing becomes bilateral at the trapezoid body, so one-sided deafness localizes to the ear, nerve, or cochlear nucleus.

10. C. Presynaptic voltage-gated calcium channels

Physiology · Synapses and neurotransmitters · Step 2

Lambert-Eaton syndrome targets presynaptic voltage-gated calcium channels, reducing ACh release, and causes proximal weakness, autonomic dysfunction, and reduced reflexes that improve with repeated contraction, often with small cell lung cancer.

E. Myasthenia gravis targets postsynaptic AChR and causes fatigable weakness.

11. D. JC virus status

Multiple sclerosis · Step 2

Natalizumab can cause PML, so JC virus status is checked first.

E. Bradycardia on starting is a concern with fingolimod, through S1P receptors on the SA and AV nodes.

12. A. Metoclopramide

Typical Parkinson disease · Step 2

Drug-induced parkinsonism follows D2 receptor blockade in the striatum, by antipsychotics and by metoclopramide and prochlorperazine. Its signs are symmetric and widespread after starting the drug, and DAT SPECT is normal.

13. E. Astrocyte

Physiology · Neurons and action potentials · Step 1

Astrocytes buffer potassium, take up neurotransmitter, and with their foot processes help form the BBB. Microglia are the immune cells, oligodendrocytes myelinate CNS axons, and ependymal cells line the ventricles.

14. E. Interferon beta

NMOSD · Step 2

Because each relapse adds disability, long-term immunosuppression starts after the first attack, with RTX, inebilizumab, IL-6 receptor blockers, eculizumab, or AZA with prednisone. Interferon beta, natalizumab, and other MS drugs can worsen NMOSD.

15. A. Pick bodies

Frontotemporal dementia · Step 1

FTLD-tau includes tau inclusions such as Pick bodies, with ballooned neurons, and knife-edge atrophy of the frontal and anterior temporal lobes.

B. Glial cytoplasmic inclusions of α-synuclein mark MSA.

16. B. Charcot-Bouchard aneurysms in penetrating arterioles

Intracerebral hemorrhage · Step 1

Chronic hypertension causes lipohyalinosis and Charcot-Bouchard aneurysms in penetrating arterioles, which bleed in the basal ganglia, thalamus, pons, and cerebellum.

A. Saccular aneurysms cause subarachnoid hemorrhage.

E. Amyloid angiopathy causes lobar bleeds.

17. A. Tetrabenazine

Physiology · Basal ganglia pathways · Step 1

VMAT2 inhibitors such as tetrabenazine block vesicular loading of dopamine, reducing chorea. Haloperidol, a D2 antagonist, also reduces hyperkinesia but can cause parkinsonism.

18. B. Pain and temperature in the left leg

Physiology · Ascending tracts · Step 1

In Brown-Séquard syndrome, the dorsal columns and corticospinal tract are lost on the same side, and spinothalamic fibers, which cross within 1–2 segments of entry, are lost on the opposite side.

19. E. Second-order sympathetic neurons leaving at T1

Physiology · III, IV, and VI · Step 1

Sympathetic fibers pass from the hypothalamus to C8–T2, out at T1 to the superior cervical ganglion, and along the internal carotid artery. A Pancoast tumor at the lung apex damages the second-order neurons, causing Horner syndrome.

20. B. Oral prednisone 1 mg/kg for 7 days, then tapered

Hearing loss · Step 2

Sudden SNHL, a loss of over 30 dB at 3 consecutive frequencies within 72 hours, is an emergency treated with oral prednisone, because spontaneous recovery is most likely in the first 2 weeks. MRI of the internal auditory canals is also obtained.

21. E. The eyes drift slowly toward the weaker side, and fast phases beat away from it

Vertigo · Step 1

Unequal signals from the two vestibular systems are read as rotation. The VOR drives a slow drift toward the weaker side, and corrective fast phases beat away from it. Nystagmus is named for its fast phase and strengthens with gaze toward that phase.

22. D. Duloxetine

Physiology · Pain pathways and descending modulation · Step 2

SNRIs and TCAs, such as duloxetine and amitriptyline, block serotonin and norepinephrine reuptake in the spinal cord, strengthening descending inhibition and relieving neuropathic pain.

23. C. Multiple system atrophy

Atypical parkinsonism · Step 1

MSA is an α-synucleinopathy with GCIs in oligodendrocytes. In PD and DLB, α-synuclein collects in neurons, and PSP and CBD are 4R tauopathies.

24. B. His INR

Acute ischemic stroke · Step 2

Contraindications to thrombolysis include INR over 1.7, platelets under 100000/µL, hemorrhage on CT, and recent DOAC use with possible ongoing effect. A BP of 170/95 is below the 185/110 threshold.

A. BP must be below 185/110, which his is.

C. He is within 4.5 hours of onset.

25. C. Mood and suicidal thoughts

Huntington disease · Step 2

Tetrabenazine inhibits VMAT2, reducing dopamine release and lessening chorea. It can cause depression, suicidality, fatigue, and akathisia, so mood and suicidal thoughts are monitored.

26. D. A DOAC

Acute ischemic stroke · Step 2

Cardiac embolus is treated with anticoagulation: a DOAC, or warfarin for a mechanical valve. Her CHA_2DS_2-VASc score is at least 5 (age 75 or older, prior stroke, female sex).

B. Endarterectomy is for symptomatic ICA stenosis of 70–99%.

C. Warfarin is chosen for a mechanical valve, which she does not have.

27. A. AQP4-IgG

Ruling out mimics · Step 2

Severe or bilateral optic neuritis, LETM, or intractable hiccups and vomiting point away from MS toward NMOSD, tested by AQP4-IgG, and some MS drugs worsen NMOSD.

28. B. 100% oxygen

Localizing a lesion · Step 2

One-sided orbital pain lasting 15–180 minutes, with tearing or nasal congestion on that side, restlessness, and night attacks, is cluster headache, relieved by 100% oxygen within 10 minutes in at least two thirds.

29. D. Foramen ovale

Physiology · V and VII · Step 1

V1 passes through the superior orbital fissure, V2 through the foramen rotundum, and V3, which supplies the lower jaw and muscles of mastication, through the foramen ovale.

30. B. Retinal or optic nerve disease in addition to the cataract

Cataract · Step 2

A cataract alone does not cause an RAPD, so an RAPD with a cataract points to retinal or optic nerve disease.

31. E. Wallerian degeneration and chromatolysis

Physiology · Neurons and action potentials · Step 1

After an axon is cut, Wallerian degeneration breaks down the axon and myelin distal to the cut, and chromatolysis swells the cell body and breaks down Nissl bodies. Regrowth, at about 1 mm/day, begins after degeneration is complete.

32. E. It blocks peripheral conversion of levodopa to dopamine

Typical Parkinson disease · Step 1

Levodopa crosses the BBB and becomes dopamine, and carbidopa blocks its peripheral conversion; together they are the most effective treatment.

C. Rasagiline is the MAO-B inhibitor.

D. Entacapone is the COMT inhibitor.

33. B. Left CN XI

Physiology · IX, X, XI, and XII · Step 1

CN XI supplies the sternocleidomastoid and trapezius and can be injured in posterior triangle node biopsy.

34. E. Triphasic waves

Choosing imaging and EEG · Step 1

Triphasic waves suggest a metabolic encephalopathy, such as hepatic or uremic encephalopathy. Prominent posterior slow waves suggest dementia with Lewy bodies.

35. A. Rods

Vision loss and red eye · Step 1

In retinitis pigmentosa, mutations kill rods first, then cones, so night blindness comes first, then tunnel vision, with central vision lost late.

36. D. Temporal lobe

Seizures · Step 1

Focal autonomic and psychological seizures arise in the temporal lobe and insula, with auras such as déjà vu and nausea. Involvement of the reticular formation or thalamus brings LOC, automatisms such as chewing and lip smacking, and a post-ictal state.

A. Frontal lobe seizures are motor.

E. An aura shows a focal onset; absence seizures have no post-ictal state.

37. C. A severe antipsychotic sensitivity reaction

Dementia with Lewy bodies · Step 2

Antipsychotics are avoided whenever possible in DLB, because serious reactions, with sudden change in consciousness, impaired swallowing, acute confusion, and worse parkinsonism, occur in up to 50%. If one is essential, low-dose quetiapine is used.

38. C. Corticospinal fibers cross at the pyramidal decussation in the medulla

Physiology · Descending tracts · Step 1

About 90% of corticospinal fibers cross at the pyramidal decussation, so lesions above it cause opposite-side weakness and cord lesions below it cause same-side weakness.

39. D. Donepezil

Alzheimer disease · Step 1

Acetylcholinesterase inhibitors slow ACh breakdown and can cause bradycardia and GI upset.

B. Lecanemab needs monitoring for ARIA.

40. D. MOGAD has no progressive phase

MOGAD · Step 1

MOGAD is monophasic or relapsing without a progressive phase, so disability comes from attacks, whereas MS can be progressive. MOGAD affects both sexes equally and is not linked to other autoimmune diseases.

C. About half of patients relapse.

41. E. Weber syndrome

Stroke symptoms and complications · Step 1

All three midbrain syndromes share CN III palsy. Contralateral hemiparesis alone means Weber syndrome (ventromedial midbrain), contralateral ataxia means Claude, and tremor with sensory loss means Benedikt.

A. Benedikt syndrome adds contralateral tremor, ataxia, and sensory loss.

B. Claude syndrome causes contralateral ataxia without weakness.

D. Millard-Gubler syndrome is pontine, with CN VI and CN VII palsy.

42. C. Left cortex or corticobulbar tract

Motor signs and syndromes · Step 1

Supranuclear facial palsy, from a contralateral cortical or corticobulbar lesion, weakens the lower face and spares the forehead, which receives input from both hemispheres. Nuclear and peripheral palsy weakens the whole half of the face.

43. A. MOG-IgG

MOGAD · Step 2

Bilateral optic neuritis with disc edema, an optic nerve lesion over half its length, and perineural optic sheath enhancement are supporting features of MOGAD, which affects men and women equally.

D. NMOSD favors posterior and chiasmal lesions, mostly in women.

44. C. Anticoagulation

Intracerebral hemorrhage · Step 2

Cerebral venous sinus thrombosis obstructs venous drainage, causing venous infarction or hemorrhage near the cortex. It is treated with anticoagulation, even with hemorrhage.

D. Hemorrhage does not prevent anticoagulation in venous sinus thrombosis.

45. C. Anterior cord syndrome

Stroke symptoms and complications · Step 1

Anterior cord syndrome, from the anterior spinal artery, causes bilateral weakness and pain and temperature loss below the lesion, preserved vibration and proprioception, and bladder and bowel dysfunction.

A. Brown-Séquard syndrome is unilateral, with contralateral pain and temperature loss.

B. Central cord syndrome affects the arms more than the legs.

D. Posterior cord lesions lose vibration and proprioception but preserve strength.

46. C. Slow circulation delays the blood gas signal to the chemoreceptors

Physiology · Respiratory center · Step 1

Cheyne-Stokes breathing alternates crescendo-decrescendo ventilation with central apnea. In heart failure, prolonged circulation time delays the chemoreceptor signal, so corrections lag behind the true blood gases.

A. Obstruction causes obstructive, not central, apnea.

B. Kussmaul breathing is continuous, without apneas.

47. A. Check and relieve bladder distension

Physiology · Cardiovascular center · Step 2

Autonomic dysreflexia follows SCI at or above T6. A stimulus below the lesion, usually a full bladder (85%), drives splanchnic vasoconstriction, while vagal bradycardia and flushing occur only above the lesion. Fixing the catheter or bladder is often life-saving.

48. E. Timolol

Glaucoma · Step 1

β-blockers such as timolol block ciliary β receptors to decrease aqueous production and are avoided in asthma, COPD, heart block, and bradycardia.

C. Latanoprost lengthens eyelashes and darkens the iris.

D. Brimonidine can cause apnea in children.

49. D. Optic chiasm

Localizing a lesion · Step 1

Bitemporal hemianopia localizes to the optic chiasm, as with a pituitary tumor.

50. E. Uhthoff phenomenon

Multiple sclerosis · Step 1

In Uhthoff phenomenon, symptoms worsen with heat. It is managed by avoiding heat and is not a relapse, which develops over days to weeks.

A. Lhermitte sign is an electric shock down the spine on neck flexion.

B. A relapse develops over days to weeks and does not settle within an hour.

Test B answers and explanations

1 **A**	2 **B**	3 **C**	4 **D**	5 **B**	6 **E**	7 **C**	8 **C**	9 **D**	10 **B**
11 **B**	12 **B**	13 **A**	14 **C**	15 **B**	16 **C**	17 **C**	18 **D**	19 **A**	20 **D**
21 **E**	22 **E**	23 **E**	24 **A**	25 **A**	26 **E**	27 **D**	28 **E**	29 **C**	30 **A**
31 **C**	32 **B**	33 **D**	34 **C**	35 **E**	36 **D**	37 **B**	38 **E**	39 **D**	40 **E**
41 **E**	42 **A**	43 **B**	44 **A**	45 **C**	46 **B**	47 **D**	48 **D**	49 **A**	50 **A**

1. A. μ receptors at the preBötzinger complex

Physiology · Respiratory center · Step 2

In opioid overdose, μ receptors at the preBötzinger complex, which generates the rhythm, blunt the response to CO_2 and hypoxia, causing pinpoint pupils, respiratory depression, and decreased consciousness. Opioids shift the CO_2 response curve right.

2. B. Meyer loop

Physiology · I, II, and VIII · Step 1

Radiations reach V1, the lower field fibers passing through the parietal lobe and the upper field through Meyer loop in the temporal lobe, so Meyer loop lesions cause an upper quadrantanopia on the opposite side.

3. C. Relapses increase during pregnancy and after delivery

NMOSD · Step 2

Unlike in MS, NMOSD relapses increase during pregnancy and after delivery. RTX has been continued safely in case reports.

4. D. T10

Physiology · Ascending tracts · Step 1

Dermatome landmarks localize sensory levels: C6 thumb, C7 middle finger, T4 nipple, T10 umbilicus, L4 anterior knee and medial malleolus, and S1 lateral malleolus.

5. B. An SSRI

Frontotemporal dementia · Step 2

SSRIs can improve behavior but not cognition. Cholinesterase inhibitors and memantine have no proven benefit, and donepezil can worsen behavior. Patients are prone to extrapyramidal effects from antipsychotics.

A. Donepezil has no proven benefit in FTD and can worsen behavior.

6. E. Noise

Hearing loss · Step 1

Noise damages hair cell stereocilia and gives a notch at 3–6 kHz, often 4 kHz, with recovery at 8 kHz. Presbycusis gives a bilateral high-frequency downsloping curve, and Ménière disease low-frequency loss.

7. C. Blocks voltage-gated sodium channels in the nerve

Physiology · Neurons and action potentials · Step 1

Local anesthetics such as lidocaine block voltage-gated sodium channels in nerves, so pain signals fail to conduct. Phenytoin, carbamazepine, and lamotrigine also reduce sodium influx.

8. C. The outermost layer of myelin and the oligodendrocyte surface

MOGAD · Step 1

MOG is a transmembrane glycoprotein of the outermost myelin lamellae and oligodendrocyte surface, so antibodies reach it easily. MOGAD is a demyelinating disease, whereas AQP4-positive NMOSD is an astrocytopathy.

9. D. D1 neurons inhibit the GPi, releasing the thalamus and allowing movement

Physiology · Basal ganglia pathways · Step 1

In the direct pathway, D1 neurons inhibit the GPi and SNr, releasing the VA and VL thalamus and allowing movement. In the indirect pathway, D2 neurons inhibit the GPe, freeing the STN to excite the GPi and restrain movement.

10. B. Common peroneal nerve at the fibular head

Motor signs and syndromes · Step 1

Common peroneal compression at the fibular head causes foot drop with weak dorsiflexion and eversion and a numb dorsum. Inversion, supplied by the tibial nerve, is spared, which separates it from L5 radiculopathy, the most common cause of foot drop.

E. An L5 lesion also weakens inversion.

11. B. A venous thrombus crossing a right-to-left shunt

Acute ischemic stroke · Step 1

Paradoxical embolus occurs with ASD, PFO, and atrial septal aneurysm, where a right-to-left shunt lets a venous thrombus reach the arterial circulation. With no other cause and age under 60, PFO closure is considered.

12. B. It was taken up by DAT and impaired mitochondria

Typical Parkinson disease · Step 1

MPTP is taken up by DAT into dopamine neurons, where it accumulates in mitochondria and impairs them, killing SNc neurons.

13. A. Reduced penetrance range

Huntington disease · Step 1

Repeats of 26 or fewer are normal, 27–35 are unstable but cause no disease, 36–39 have reduced penetrance, 40 or more almost always cause disease, and over 55–60 cause juvenile disease.

14. C. Glucose and non-contrast CT head

Acute ischemic stroke · Step 2

Glucose and non-contrast CT head come first, to rule out hypoglycemia and hemorrhage before thrombolysis. If there is no hemorrhage, CT angiogram looks for large vessel occlusion.

B. Aspirin is given within 24–48 hours, after hemorrhage is excluded.

D. For thrombolysis, BP needs only to be below 185/110.

E. Hemorrhage must be excluded before thrombolysis.

15. B. The Epley maneuver

Vertigo · Step 2

Posterior canal BPPV causes episodes under 1 minute with upbeat, torsional nystagmus on the Dix-Hallpike maneuver, and is treated with the Epley maneuver, which repositions otoconia out of the posterior canal.

D. Suppressants beyond a few days delay central compensation.

16. C. 4

Intracerebral hemorrhage · Step 2

The ICH score gives 1 point for GCS 5–12 (2 for 3–4), 1 for age 80 or older, 1 for infratentorial origin, 1 for volume of 30 mL or more, and 1 for IVH. He scores 1 + 1 + 0 + 1 + 1 = 4. The score estimates 30-day mortality.

17. C. Decreased Aβ42 with increased tau and phosphorylated tau

Alzheimer disease · Step 2

CSF in Alzheimer disease shows decreased Aβ42 with increased tau and phosphorylated tau, and amyloid PET shows plaques in life.

18. D. An extra copy of the APP gene on chromosome 21

Alzheimer disease · Step 1

In Down syndrome, chromosome 21 carries an extra APP gene, increasing Aβ production.

19. A. Vasospasm and delayed cerebral ischemia

Stroke symptoms and complications · Step 2

Subarachnoid hemorrhage is complicated by rebleeding, highest early before the aneurysm is secured, and by vasospasm and delayed cerebral ischemia on days 3–14. New focal deficits on day 6 without new blood fit vasospasm.

B. Hemorrhagic transformation follows ischemic stroke and would show blood within an infarct.

E. Her aneurysm is secured, and CT shows no new blood.

20. D. NTS excites the CVLM, which inhibits the RVLM

Physiology · Cardiovascular center · Step 1

Baroreceptor afferents release glutamate in the NTS, which excites the CVLM, whose GABA neurons inhibit the RVLM, the source of sympathetic drive. The NTS also excites cardiac vagal neurons in the nucleus ambiguus.

21. E. GABA-A receptor, chloride entry

Physiology · Synapses and neurotransmitters · Step 1

Inhibitory postsynaptic potentials hyperpolarize the next neuron, as chloride enters through GABA-A receptors. Inputs sum at the axon hillock.

22. E. Nuclear sclerosis

Cataract · Step 1

Nuclear sclerosis, the most common type, turns the center yellow-brown and causes a myopic shift that lets some patients read again without glasses.

A. Posterior subcapsular cataract causes glare and affects vision most.

C. Snowflake cortical opacities occur in diabetes.

23. E. IV nicardipine to a systolic target of 140

Intracerebral hemorrhage · Step 2

If systolic is 150–220, it is lowered to a target of 140 with IV nicardipine, clevidipine, or labetalol, to limit hematoma growth.

A. Thrombolysis is contraindicated by hemorrhage.

C. Permissive hypertension is for ischemic stroke.

24. A. Subarachnoid hemorrhage from a saccular aneurysm

Subarachnoid hemorrhage · Step 1

PKD is one of the inherited conditions, with Ehlers-Danlos syndrome, fibromuscular dysplasia, coarctation of the aorta, and Marfan syndrome, linked with aneurysms, and these conditions need screening.

25. A. The α2δ-1 subunit of voltage-gated calcium channels

Physiology · Pain pathways and descending modulation · Step 1

Gabapentin and pregabalin bind the α2δ-1 subunit of voltage-gated calcium channels, reducing calcium entry and excitatory transmitter release, which relieves neuropathic pain.

26. E. Corneal scraping, then urgent topical antibiotics

Vision loss and red eye · Step 2

Bacterial keratitis is a sight-threatening corneal infection, most often after contact lens wear, from organisms such as Staphylococcus aureus and Pseudomonas. Corneal scraping for Gram stain and culture precedes urgent topical antibiotics such as moxifloxacin.

C. Supportive care is for viral conjunctivitis, which spares vision.

27. D. Drug-induced parkinsonism

Ruling out mimics · Step 1

DAT SPECT is reduced in Parkinson disease and atypical parkinsonism alike, so it separates them from drug-induced parkinsonism and essential tremor, which have normal uptake, not from each other.

28. E. Urgent surgical repair

Vision loss and red eye · Step 2

Retinal detachment causes floaters and flashes, then field loss spreading like a curtain. Urgent surgery seals the break, and the outcome is best if the macula is still attached.

29. C. Medial longitudinal fasciculus

Multiple sclerosis · Step 1

MLF damage causes INO, which is bilateral in multiple sclerosis: the adducting eye fails while the abducting eye has nystagmus, with convergence preserved.

E. Oculomotor palsy would also impair convergence and cause ptosis.

30. A. Blood blocks CSF absorption at the arachnoid granulations

Subarachnoid hemorrhage · Step 1

Blood blocks CSF absorption at the arachnoid granulations, causing hydrocephalus, which is treated with an EVD.

31. C. Unopposed vestibulospinal and pontine reticulospinal tracts

Physiology · Descending tracts · Step 1

The lateral vestibulospinal and pontine reticulospinal tracts excite antigravity extensors. A lesion below the red nucleus removes rubrospinal flexion, causing decerebrate posturing with extension of all limbs.

D. The rubrospinal tract favors upper limb flexion; preserved above the lesion, it causes decorticate posturing.

32. B. It does not exclude epilepsy

Ruling out mimics · Step 1

A normal EEG during a typical event points to PNES, whereas a normal routine EEG between events does not exclude epilepsy.

33. D. Less salt, caffeine, and alcohol, with a thiazide or betahistine

Vertigo · Step 2

Ménière disease, from endolymphatic hydrops, causes attacks of 20 minutes to 12 hours with fluctuating low-frequency loss, tinnitus, and fullness. It is treated with less salt, caffeine, and alcohol, thiazides or betahistine, intratympanic steroids, and intratympanic gentamicin if refractory.

34. C. GJB2

Hearing loss · Step 1

Nonsyndromic loss is autosomal recessive in 75–80%, so parents are usually unaffected carriers. DFNB1 mutations in GJB2 and GJB6, encoding connexins 26 and 30, are the most common genetic cause of severe to profound congenital deafness.

B. PAX3 causes Waardenburg syndrome types 1 and 3, with pigment changes.

E. Merlin mutations cause NF2, with bilateral vestibular schwannomas.

35. E. Relative preservation of the medial temporal lobe

Dementia with Lewy bodies · Step 1

In DLB, the medial temporal lobe is relatively preserved, so memory is spared early, while cortical involvement causes attention, executive, and visuospatial decline.

36. D. Laser peripheral iridotomy

Glaucoma · Step 2

Acute angle closure is lowered with acetazolamide, timolol, and apraclonidine, then IV mannitol if needed and pilocarpine once IOP is below 40 mmHg. Laser peripheral iridotomy, which lets aqueous humor bypass the pupil, is the treatment of choice.

C. Atropine dilates the pupil and can trigger or worsen angle closure.

E. Laser trabeculoplasty treats open-angle glaucoma.

37. B. Lumbar puncture with opening pressure

Localizing a lesion · Step 2

Papilledema, pulsatile tinnitus, and CN VI palsy in a young woman with obesity suggest IIH once a mass is excluded, and call for imaging and then a lumbar puncture with opening pressure.

38. E. Left CN X

Physiology · IX, X, XI, and XII · Step 1

The uvula deviates away from a vagal lesion, so deviation to the right means a left CN X lesion. The tongue deviates toward a hypoglossal lesion.

39. D. Observation, expecting it to resolve

Seizures · Step 2

Todd paralysis is temporary weakness after a focal or generalized tonic-clonic seizure, lasting minutes to 1–2 days and resolving without lasting damage. Mistaking it for a stroke can lead to unneeded thrombolysis.

E. A witnessed seizure with tongue biting points to Todd paralysis, a stroke mimic.

40. E. Ventilation-perfusion mismatch and the Haldane effect

Physiology · Respiratory center · Step 2

Oxygen raises CO_2 in COPD through ventilation-perfusion mismatch, as hypoxic vasoconstriction is reversed, and the Haldane effect, rather than loss of hypoxic drive, so saturations of 88–92% are targeted.

41. E. It increases, inhibiting the thalamus and causing bradykinesia

Physiology · Basal ganglia pathways · Step 1

Loss of SNc dopamine weakens the direct pathway and strengthens the indirect pathway, increasing GPi inhibition of the thalamus and causing bradykinesia.

42. A. Varicella zoster of the facial nerve

Physiology · V and VII · Step 2

Ramsay Hunt syndrome, varicella zoster of the facial nerve with a vesicular rash, causes about 6% of facial palsy.

43. B. Its long course up the clivus makes it vulnerable to raised ICP

Physiology · III, IV, and VI · Step 1

CN VI arises under the facial colliculus, climbs the clivus, and runs free in the cavernous sinus; its palsy causes esotropia and can be a false localizing sign of raised ICP.

D. CN IV is the only nerve to leave the back of the brainstem.

44. A. Ethosuximide

Seizures · Step 2

Absence seizures, usually at ages 5–15, cause staring and automatisms with no post-ictal state, and EEG shows a 3 Hz spike-and-wave pattern. Ethosuximide, which blocks T-type calcium channels, treats them.

B. Vigabatrin is for West syndrome.

C. Carbamazepine is for focal seizures.

45. C. Propranolol

Motor signs and syndromes · Step 2

Postural tremor of both hands at 6–12 Hz, often familial, is essential tremor, treated with propranolol or primidone.

46. B. Risk of NSF

Choosing imaging and EEG · Step 2

Gadolinium is avoided in severe renal impairment, because of NSF, and where possible in pregnancy.

47. D. SNARE-mediated vesicle fusion

Physiology · Synapses and neurotransmitters · Step 1

Calcium entry through voltage-gated calcium channels activates SNARE proteins, which fuse vesicles with the membrane and release transmitter.

48. D. Midodrine

Atypical parkinsonism · Step 2

Orthostatic hypotension in atypical parkinsonism is treated with midodrine or fludrocortisone.

E. α-blockers treat urinary dysfunction and can worsen orthostatic hypotension.

49. A. Indirect-pathway medium spiny neurons projecting to the GPe

Huntington disease · Step 1

Indirect-pathway neurons, which contain enkephalin and project to the GPe, are lost first, so the GPi inhibits the thalamus less and the released thalamus overdrives the cortex, causing chorea. Direct-pathway loss later causes rigidity and bradykinesia.

50. **A. Behavioral variant frontotemporal dementia**

Frontotemporal dementia · Step 2

bvFTD causes disinhibition, apathy, loss of empathy, compulsive behavior, hyperorality, and executive dysfunction with anosognosia, while memory and visuospatial skills are spared early. Possible bvFTD needs 3 of the 6 behavioral features.

Test C answers and explanations

1 **A**	2 **A**	3 **B**	4 **A**	5 **B**	6 **D**	7 **B**	8 **E**	9 **E**	10 **D**
11 **A**	12 **C**	13 **A**	14 **E**	15 **A**	16 **E**	17 **B**	18 **B**	19 **D**	20 **C**
21 **E**	22 **C**	23 **B**	24 **C**	25 **A**	26 **B**	27 **A**	28 **D**	29 **A**	30 **C**
31 **D**	32 **D**	33 **C**	34 **E**	35 **D**	36 **C**	37 **E**	38 **E**	39 **C**	40 **D**
41 **C**	42 **B**	43 **E**	44 **B**	45 **B**	46 **A**	47 **E**	48 **D**	49 **C**	50 **D**

1. A. Caudate atrophy

Huntington disease · Step 1

MRI in Huntington disease shows early caudate atrophy, even before symptoms, with cortical and cerebellar atrophy later. Caudate atrophy tracks repeat length and motor score.

2. A. CSF oligoclonal bands

Multiple sclerosis · Step 2

The 2024 criteria need dissemination in space, with plaques in at least 2 of the 5 sites, here met with the optic nerve, periventricular, and juxtacortical sites. Dissemination in time is missing, but CSF oligoclonal bands, kappa free light chains, 4 or more sites, or 6 plaques with a central vein sign can substitute for time.

B. Positive AQP4 antibody points to NMOSD rather than MS.

3. B. It excites D1 neurons and inhibits D2 neurons

Physiology · Basal ganglia pathways · Step 1

Dopamine excites D1 neurons, strengthening the direct pathway, and inhibits D2 neurons, weakening the indirect pathway, so both actions favor movement.

4. A. Dementia with Lewy bodies

Dementia with Lewy bodies · Step 2

The four core features of DLB are fluctuating cognition, recurrent well-formed visual hallucinations, RBD, and parkinsonism; probable DLB needs 2 or more. DLB is diagnosed when dementia begins before, with, or within 1 year of parkinsonism.

E. PDD is diagnosed when dementia begins 1 year or more after established Parkinson disease.

5. B. Pyridostigmine

Physiology · Synapses and neurotransmitters · Step 2

Myasthenia gravis targets postsynaptic nicotinic AChR or MuSK, causing fatigable weakness with ptosis and diplopia in 85%. Pyridostigmine, a cholinesterase inhibitor, comes first, then immunosuppression.

6. D. Gonioscopy

Glaucoma · Step 2

Tonometry measures IOP, gonioscopy shows whether the angle is open or closed, fundoscopy and OCT assess the cup and nerve fiber layer, and visual fields show peripheral and arcuate loss.

7. B. Left CN XII

Physiology · IX, X, XI, and XII · Step 1

A CN XII lesion deviates the tongue toward the lesion, with wasting and fasciculations.

8. E. Nerve to stapedius

Physiology · V and VII · Step 1

In the facial canal, CN VII supplies the stapedius and forms the chorda tympani, which carries taste from the front two-thirds of the tongue. Loss of the stapedius causes hyperacusis.

9. E. Increases uveoscleral outflow

Glaucoma · Step 1

Prostaglandin analogs such as latanoprost increase uveoscleral outflow and are first line; they lengthen eyelashes and darken the iris.

B. This is how carbonic anhydrase inhibitors work.

C. This is how pilocarpine works.

D. This is how timolol works.

10. D. NMOSD

NMOSD · Step 1

CSF OCB are usually absent in NMOSD, and their presence is a red flag. Over 50 white cells/µL, or neutrophils or eosinophils, favor NMOSD over MS.

11. A. Neurosyphilis

Physiology · I, II, and VIII · Step 1

Argyll Robertson pupils constrict to a near target but not to light and mark neurosyphilis.

12. C. Right

Physiology · Ascending tracts · Step 1

Both spinocerebellar tracts end in the cerebellum on the same side, the ventral tract by crossing twice, so their lesions cause same-side ataxia, as in Friedreich ataxia.

13. A. Donepezil

Alzheimer disease · Step 2

Acetylcholinesterase inhibitors, donepezil, rivastigmine, and galantamine, offset the loss of cholinergic neurons from the nucleus basalis of Meynert and give a modest benefit in mild to moderate disease. Memantine, an NMDA antagonist, is for moderate to severe disease.

C. Memantine is for moderate to severe disease.

14. E. In the medulla, as internal arcuate fibers

Physiology · Ascending tracts · Step 1

In the dorsal column-medial lemniscus, fibers ascend on the same side to the nucleus gracilis and cuneatus, then second-order internal arcuate fibers cross in the medulla and ascend as the medial lemniscus to the VPL.

B. Spinothalamic fibers cross in the anterior white commissure.

15. A. A sympathetic surge to maintain CPP as ICP rises

Physiology · Cardiovascular center · Step 1

In the Cushing reflex, raised ICP lowers CPP, the difference between MAP and ICP, causing a sympathetic surge with widened pulse pressure, then bradycardia and irregular respirations, a late sign of herniation.

16. E. Left internal capsule

Localizing a lesion · Step 1

Equal weakness of face, arm, and leg points to the internal capsule, where the fibers are packed together. Cortical signs such as aphasia and neglect would place it in the cortex.

17. B. Facial weakness that includes the forehead points to a peripheral lesion

Acute ischemic stroke · Step 2

Bell palsy is a stroke mimic that causes isolated facial weakness including the forehead. Mimics must be recognized without delaying reperfusion.

18. B. Left pons

Localizing a lesion · Step 1

A cranial nerve palsy with weakness of the opposite body places the lesion in the brainstem at the level of that nerve. CN VI arises in the pons.

19. D. NMOSD

NMOSD · Step 1

LETM over 3 or more vertebral segments centered on the gray matter, and lesions where AQP4 is dense, such as the dorsal medulla, suggest NMOSD. Short, peripheral cord lesions favor MS.

20. C. Video EEG

Seizures · Step 1

Closed eyes, side-to-side head movement, asynchronous limb movement, and no post-ictal state suggest psychogenic non-epileptic seizures, confirmed by video EEG.

D. Prolactin also rises after convulsive syncope, and a normal level does not confirm PNES.

21. E. It causes a floppy iris during surgery

Cataract · Step 2

Before surgery, tamsulosin, which causes a floppy iris, and anticoagulants are noted.

22. C. Visceral and somatic afferents converge on the same spinal segments

Physiology · Pain pathways and descending modulation · Step 1

Referred pain arises when visceral and somatic afferents converge on the same spinal segments, so the brain misattributes visceral pain to the body wall.

23. B. HLA-B*1502

Seizures · Step 1

Carbamazepine causes SJS, agranulocytosis, aplastic anemia, and hyponatremia. Patients of Han Chinese ancestry are tested for HLA-B*1502 because it predicts SJS.

24. C. ECG and orthostatic BP

Ruling out mimics · Step 2

A trigger, lightheadedness, brief jerks, and rapid recovery suggest syncope, assessed with ECG and orthostatic BP.

25. A. Melatonin

Dementia with Lewy bodies · Step 2

RBD in DLB is treated with melatonin, or cautious bedtime clonazepam.

26. B. Axial and proximal muscles on both sides

Physiology · Descending tracts · Step 1

The anterior corticospinal tract, about 10%, stays uncrossed until its segment and serves axial and proximal muscles on both sides, whereas the lateral tract drives skilled distal movement.

27. A. Paracetamol

Subarachnoid hemorrhage · Step 2

Until the aneurysm is secured, antithrombotics are stopped, anticoagulation is reversed, and pain is treated with paracetamol, avoiding aspirin and NSAIDs.

28. D. C9orf72 repeat expansion

Frontotemporal dementia · Step 1

The C9orf72 repeat expansion is the most common cause of familial FTD and ALS, and causes FTLD-TDP; TDP-43 is also the protein of ALS.

A. MAPT causes FTLD-tau.

29. A. IVIG every 4 weeks

MOGAD · Step 2

Relapse prevention usually starts after a second attack, with AZA or MMF bridged by prednisolone, IVIG at 1–2 g/kg every 4 weeks, or oral corticosteroids. MS drugs such as interferons, glatiramer acetate, and fingolimod appear ineffective.

B. About half of patients have a single attack, but he has relapsed.

30. C. Retained bicarbonate blunts the central chemoreceptors

Physiology · Respiratory center · Step 1

In obesity hypoventilation, chest wall mechanics cause hypoventilation, and the kidneys retain bicarbonate, which blunts the sensitivity of the central chemoreceptors, so hypercapnia persists.

31. D. Potassium leak channels make the membrane most permeable to potassium at rest

Physiology · Neurons and action potentials · Step 1

Potassium leak channels make the resting membrane most permeable to potassium, so the resting potential lies near the potassium equilibrium potential. The Na/K-ATPase pumps 3 sodium out and 2 potassium in to maintain the gradients.

B. The pump moves potassium into the cell.

32. D. Left posterior cerebral artery

Stroke symptoms and complications · Step 1

PCA syndrome usually causes no weakness, with contralateral homonymous hemianopia with macular sparing. Alexia without agraphia occurs with dominant (left) lesions, and prosopagnosia with non-dominant lesions.

B. MCA syndrome causes face and arm weakness.

E. A right PCA stroke causes a left hemianopia and prosopagnosia.

33. C. VEGF

Vision loss and red eye · Step 1

In wet macular degeneration, VEGF drives new vessels from the choroid through Bruch membrane, and they leak and bleed. Intravitreal anti-VEGF agents, ranibizumab or aflibercept, are the treatment.

D. Drusen build up beneath the RPE in dry AMD.

34. E. Permanent pacemaker

Physiology · Cardiovascular center · Step 2

Carotid sinus hypersensitivity is cardioinhibitory (asystole of 3 seconds or more) in 70%, vasodepressor in 10%, and mixed in 20%. Pacing treats the cardioinhibitory type.

35. D. MRI with SWI

Choosing imaging and EEG · Step 2

SWI shows blood products as dark, revealing the lobar microbleeds of amyloid angiopathy.

36. C. Failed migration of neural crest cells

Hearing loss · Step 1

Waardenburg syndrome follows failed migration of neural crest cells, so melanocytes are missing. Type 1, from PAX3, is autosomal dominant with a white forelock, heterochromia, and dystopia canthorum.

37. E. Vestibular neuritis

Vertigo · Step 1

Vestibular neuritis, presumed viral inflammation of the vestibular nerve, causes acute continuous vertigo with peripheral HINTS findings and normal hearing. Labyrinthitis adds unilateral hearing loss and tinnitus.

38. E. Nimodipine

Subarachnoid hemorrhage · Step 2

Blood around arteries narrows them on days 3–14, causing vasospasm and delayed cerebral ischemia. Nimodipine prevents ischemia, and refractory vasospasm is treated with euvolemia and endovascular therapy.

C. Aspirin and NSAIDs are avoided until the aneurysm is secured.

D. An antifibrinolytic may be given only if repair is delayed, for no more than 72 hours.

39. C. It reaches cortex without a thalamic relay

Physiology · I, II, and VIII · Step 1

Olfactory receptor neurons, which regenerate for life, pass through the cribriform plate to mitral cells in the olfactory bulb, which project to piriform cortex, amygdala, and entorhinal cortex without a thalamic relay.

40. D. At or below the red nucleus, in the midbrain or rostral pons

Motor signs and syndromes · Step 1

Decorticate posturing, with flexed arms and extended legs, follows lesions above the red nucleus, and decerebrate posturing, with extension of all limbs, follows lesions at or below it.

41. C. Right subthalamic nucleus

Motor signs and syndromes · Step 1

Hemichorea and hemiballismus classically follow a lesion of the contralateral subthalamic nucleus, most often from ischemic stroke over 65, and tend to resolve.

42. B. IV methylprednisolone for 3–7 days

Multiple sclerosis · Step 2

An acute relapse is treated with IV methylprednisolone for 3–7 days, with plasma exchange if refractory.

A. Plasma exchange is for relapses refractory to steroids.

E. Baclofen treats spasticity.

43. E. MRI with gadolinium of the internal auditory canals

Hearing loss · Step 2

Asymmetric SNHL or unilateral tinnitus calls for MRI with gadolinium to look for vestibular schwannoma, a benign Schwann cell tumor of CN VIII.

D. Presbycusis is bilateral and symmetric.

44. B. Normal striatal uptake on DAT SPECT

Typical Parkinson disease · Step 2

DAT SPECT uptake is reduced with dopamine loss, in PD, MSA, PSP, CBD, and DLB, and normal in essential tremor and drug-induced and vascular parkinsonism. Normal DAT imaging is an absolute exclusion criterion for PD.

45. B. Right MLF

Physiology · III, IV, and VI · Step 1

The MLF links the CN VI nucleus to the opposite CN III nucleus. A right MLF lesion stops the right medial rectus from following on left gaze, causing INO, as in MS.

46. A. Left anterior cerebral artery

Stroke symptoms and complications · Step 1

ACA syndrome causes contralateral leg weakness greater than arm, contralateral leg sensory loss, urinary incontinence, abulia, and a grasp reflex. Right-sided signs place it in the left hemisphere.

B. The deficits are on the right, so the lesion is on the left.

C. MCA syndrome favors the face and arm over the leg.

E. Lacunar infarcts lack cortical signs such as abulia and a grasp reflex.

47. E. Cerebral amyloid angiopathy

Intracerebral hemorrhage · Step 1

Cerebral amyloid angiopathy is a common cause of lobar bleeds in older adults, may present with dementia, and shows lobar microbleeds on MRI with SWI. Hypertension bleeds in deep structures such as the basal ganglia, thalamus, pons, and cerebellum.

B. AVMs are a cause in young patients.

C. Hypertensive bleeds are deep, and she has no hypertension.

48. D. Thrombus in the artery, highly specific for LVO

Choosing imaging and EEG · Step 1

A hyperdense MCA sign reflects thrombus in the artery and is highly specific for LVO. Early ischemia otherwise often looks normal on CT.

49. C. Reduce or stop pramipexole

Typical Parkinson disease · Step 2

Dopamine agonists such as pramipexole are the main trigger of impulse control disorders, including gambling, hypersexuality, compulsive shopping, and binge eating, which call for reducing or stopping the agonist.

50. D. Hot cross bun sign in the pons

Atypical parkinsonism · Step 2

Early autonomic failure, cerebellar or pyramidal signs, and stridor favor MSA, in which MRI shows the hot cross bun sign in the pons, atrophy of the putamen, MCP, pons, and cerebellum, and putaminal iron.

A. The hummingbird sign marks PSP.

C. Asymmetric frontoparietal atrophy marks CBD.